

Super Catalan Numbers and Fourier Summation over Finite Fields

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Abstract

We study polynomial summation over unit circles over finite fields of odd characteristic, obtaining a purely algebraic integration theory without recourse to infinite procedures. There are nonetheless strong parallels to classical integration theory over a circle, and we show that the super Catalan numbers and closely related rational numbers lie at the heart of both theories. This gives a uniform analytic meaning to these up to now somewhat mysterious numbers.

Our derivation utilises the three-fold symmetry of chromogeometry between Euclidean and relativistic geometries, and we find that the Fourier summation formulas we derive in these two different settings are closely connected.

1 Introduction

This paper will elucidate the crucial and perhaps surprising role that super Catalan numbers, and closely associated rational numbers which we will call circular super Catalan numbers, play in algebra and analysis. **Super Catalan numbers** are natural numbers of the form

$$S(m, n) \equiv \frac{(2m)! (2n)!}{m! n! (m+n)!}$$

and they generalize the Catalan numbers, since $S(1, n)$ is twice the Catalan number $C(n)$. While the super Catalan numbers were first introduced by Catalan [6] in 1874, it appears that the first modern study of these numbers was initiated by Gessel [16] in 1992.

Clearly $S(m, n)$ is symmetric in m and n , and $S(0, n) = S(n, n) = \binom{2n}{n}$. The fact that $S(m, n)$ is always an integer is a simple consequence of the recurrence relation $4S(m, n) = S(m+1, n) + S(m, n+1)$. The table below lists $S(m, n)$ for $0 \leq m, n \leq 10$.

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$m \backslash n$	0	1	2	3	4	5	6	7	8	9	10
0	1	2	6	20	70	252	924	3432	12 870	48 620	184 756
1	2	2	4	10	28	84	264	858	2860	9724	33 592
2	6	4	6	12	28	72	198	572	1716	5304	16 796
3	20	10	12	20	40	90	220	572	1560	4420	12 920
4	70	28	28	40	70	140	308	728	1820	4760	12 920
5	252	84	72	90	140	252	504	1092	2520	6120	15 504
6	924	264	198	220	308	504	924	1848	3960	8976	21 318
7	3432	858	572	572	728	1092	1848	3432	6864	14 586	32 604
8	12 870	2860	1716	1560	1820	2520	3960	6864	12 870	25 740	54 340
9	48 620	9724	5304	4420	4760	6120	8976	14 586	25 740	48 620	97 240
10	184 756	33 592	16 796	12 920	12 920	15 504	21 318	32 604	54 340	97 240	184 756

Table 1: $S(m, n)$ for $0 \leq m, n \leq 10$.

Somewhat surprisingly, there is no known general combinatorial understanding for $S(m, n)$ to date, unlike the Catalan numbers (sequence A000108 in the OEIS [26]) which have hundreds of combinatorial interpretations, compiled for example in [33]. There are however some combinatorial or weighted interpretations of $S(m, n)$ for specific values of m and n ; in terms of blossom trees [30], cubic trees [27], or pairs of Dyck paths with restricted heights for $(m, n) = (2, 3)$ by Gessel [17].

Chen and Wang gave an interpretation for $S(m, m+s)$ for $0 \leq s \leq 3$ in terms of restricted lattice paths [7]. Recently, a combinatorial interpretation for $S(3, n)$ and $S(4, n)$ was given by Gheorghiciuc and Orelowitz [15] as the numbers of 3-tuples and 4-tuples of Dyck paths satisfying certain conditions.

A weighted interpretation of $S(m, n)$ involve counting 2-Motzkin paths was discovered by Allen and Gheorghiciuc [1], and another in terms of certain values of Krawtchouk polynomials was found by Georgiadis, Munemasa, and Tanaka in [14]. The latter will play a prominent role in this paper.

Finally, there are some other established identities involving the super Catalan numbers, for example relations to the reciprocal super Catalan matrix were discovered by Prodinger [28]. Liu [22] found an identity for congruences of the super Catalan numbers, and the paper by Mikić [25] gives an identity for convolution of the super Catalan numbers.

In this paper we introduce also a closely related family of rational numbers, the **circular super Catalan numbers** given by

$$\Omega(m, n) \equiv \frac{S(m, n)}{4^{m+n}}$$

and a table of $\Omega(m, n)$ for small values of m and n is given below.

$m \backslash n$	0	1	2	3	4	5	6	7	8
0	1	$\frac{1}{2}$	$\frac{3}{8}$	$\frac{5}{16}$	$\frac{35}{128}$	$\frac{63}{256}$	$\frac{231}{1024}$	$\frac{429}{2048}$	$\frac{6435}{32768}$
1	$\frac{1}{2}$	$\frac{1}{8}$	$\frac{1}{16}$	$\frac{5}{128}$	$\frac{7}{256}$	$\frac{21}{1024}$	$\frac{33}{2048}$	$\frac{429}{32768}$	$\frac{715}{65536}$
2	$\frac{3}{8}$	$\frac{1}{16}$	$\frac{3}{128}$	$\frac{3}{256}$	$\frac{7}{1024}$	$\frac{9}{2048}$	$\frac{99}{32768}$	$\frac{143}{65536}$	$\frac{429}{262144}$
3	$\frac{5}{16}$	$\frac{5}{128}$	$\frac{3}{256}$	$\frac{5}{1024}$	$\frac{5}{2048}$	$\frac{45}{32768}$	$\frac{55}{65536}$	$\frac{143}{262144}$	$\frac{195}{524288}$
4	$\frac{35}{128}$	$\frac{7}{256}$	$\frac{7}{1024}$	$\frac{5}{2048}$	$\frac{35}{32768}$	$\frac{35}{65536}$	$\frac{77}{262144}$	$\frac{91}{524288}$	$\frac{455}{4194304}$
5	$\frac{63}{256}$	$\frac{21}{1024}$	$\frac{9}{2048}$	$\frac{45}{32768}$	$\frac{35}{65536}$	$\frac{63}{262144}$	$\frac{63}{524288}$	$\frac{273}{4194304}$	$\frac{315}{8388608}$
6	$\frac{231}{1024}$	$\frac{33}{2048}$	$\frac{99}{32768}$	$\frac{55}{65536}$	$\frac{77}{262144}$	$\frac{63}{524288}$	$\frac{231}{4194304}$	$\frac{231}{8388608}$	$\frac{495}{33554432}$
7	$\frac{429}{2048}$	$\frac{429}{32768}$	$\frac{143}{65536}$	$\frac{143}{262144}$	$\frac{91}{524288}$	$\frac{273}{4194304}$	$\frac{231}{8388608}$	$\frac{429}{33554432}$	$\frac{429}{67108864}$
8	$\frac{6435}{32768}$	$\frac{715}{65536}$	$\frac{429}{262144}$	$\frac{195}{524288}$	$\frac{455}{4194304}$	$\frac{315}{8388608}$	$\frac{495}{33554432}$	$\frac{429}{67108864}$	$\frac{6435}{2147483648}$

Table 2: $\Omega(m, n)$ for $0 \leq m, n \leq 8$.

We see that the symmetry property for $\Omega(m, n)$ is still retained, and the recurrence property of the super Catalan numbers becomes the Pascal-like property

$$\Omega(m, n) = \Omega(m+1, n) + \Omega(m, n+1).$$

It is not hard to see that the table above is completely determined by the values in the zeroth row, the symmetry property, and the Pascal-like property above.

The objective of this paper is to show that the circular super Catalan numbers have a central algebraic and analytic interpretation in the context of integration, or summation, theory of polynomials over unit circles in finite fields of odd characteristic. This role extends even to summations over suitably defined unit circles in relativistic geometries, and the curious relations between the Euclidean and relativistic formulas are an aspect of the three-fold symmetry of *chromogeometry* [37].

This theory brings together Euclidean planar geometry with quadratic form $x^2 + y^2$ (blue) with two different forms of relativistic Einstein-Minkowski geometries with quadratic forms $x^2 - y^2$ (red) and xy (green). The relations and symmetries between the three geometries play a crucial role in our derivations, as summations over unit circles in both Euclidean and relativistic geometries are closely related, and connections between them provide a powerful tool for explicit evaluations.

We briefly summarize the remarkable formulas in which $\Omega(m, n)$ play the primary role and illustrate them with examples below. We will also observe that the classical formulas for integration on the circle in the Euclidean plane can, somewhat surprisingly, also be restated in terms of super Catalan numbers, so that we obtain a bridge between characteristic zero and prime characteristic harmonic analysis.

We will see that these formulas also suggest the study of families of additional rational numbers that play secondary roles.

1.1 Summing polynomials over unit circles over finite fields

The problem of summing polynomials over a finite field has many implications to number theory, Fourier analysis, and more generally harmonic analysis. In 2015, Thakur gave an expository survey

on sums of integral powers of monic polynomials of certain degrees over finite fields [36], in which relations to the zeta function also appear. For work aimed more in the direction of complex number values see [3], [4], [18], and [34].

To briefly introduce the key ideas in our approach, we begin with a finite field $\mathbb{F}_q$ with multiplicative identity 1_q , where $q = p^r$ is a power of an odd prime p , so that $\mathbb{F}_p \subseteq \mathbb{F}_q$. The associated affine plane $\mathbb{A} = \mathbb{A}(\mathbb{F}_q)$ consists of points $[x, y]$ where $x, y \in \mathbb{F}_q$. We introduce three different metrical geometries: the Euclidean and two relativistic geometries which we color-code as blue (b), red (r), and green (g) with respective associated quadratic forms $x^2 + y^2$, $x^2 - y^2$, and xy , and where the subscript c refers to any of these. The unit circles in each geometry are then defined by

$$\begin{aligned} S_b(\mathbb{F}_q) &\equiv \{[x, y] \in \mathbb{A} : x^2 + y^2 = 1\} \\ S_r(\mathbb{F}_q) &\equiv \{[x, y] \in \mathbb{A} : x^2 - y^2 = 1\} \\ S_g(\mathbb{F}_q) &\equiv \{[x, y] \in \mathbb{A} : xy = 1\}. \end{aligned}$$

We will use $S_{c,q}$ as a shorthand for $S_c(\mathbb{F}_q)$ when the underlying field is clear from context.

Recall that for a prime power $q = p^r$ the **Jacobi symbol** is defined as

$$\left(\frac{-1}{q}\right) \equiv \left(\frac{-1}{p}\right)^r$$

where the notation $\left(\frac{-1}{p}\right)$ refers to the Legendre symbol, which has value 1 if -1 is a quadratic residue mod p , that is a square in the field $\mathbb{F}_p$, and has value of -1 if -1 is not a quadratic residue. The Jacobi symbol comes into play in the following elementary count of the sizes of the unit circles, which we present in the order green, red, and blue here to align with the natural increasing complexity of formulas.

Lemma 1. *Let $S_{b,q}$, $S_{r,q}$, and $S_{g,q}$ be the unit circles in the blue, red, and green geometries respectively over the field $\mathbb{F}_q$. Then*

$$|S_{g,q}| = q - 1 \quad |S_{r,q}| = q - 1 \quad \text{and} \quad |S_{b,q}| = q - \left(\frac{-1}{q}\right).$$

Polynumbers are a variant of polynomials introduced by Wildberger in his online *Maths Foundation* lectures, and can be defined without having a prior notion of variables. This allows us to more clearly distinguish between *formal polynomials* and *polynomial functions*, and allows us to build an integration or summation theory over finite fields in a cleaner and more rigorous way. We may think of a polynumber as a list or array of coefficients, equipped with the usual Cauchy product, giving us the algebra $\text{Pol}_2(\mathbb{F}_q)$. We can evaluate a polynumber π to get a polynomial function $\varepsilon(\pi)$, so there is an evaluation map $\pi \mapsto \varepsilon(\pi)$.

For each geometry we define a linear functional on $\text{Pol}_2(\mathbb{F}_q)$, called the **Fourier summation functional**, as follows:

$$\begin{aligned} \psi_{c,q} : \text{Pol}_2(\mathbb{F}_q) &\rightarrow \mathbb{F}_q \\ \pi &\mapsto \frac{1}{|S_{c,q}|} \sum_{[x,y] \in S_{c,q}} \varepsilon(\pi)(x, y) \end{aligned}$$

where we treat $|S_{c,q}|$ as an element of $\mathbb{F}_q$. Its reciprocal is well-defined since $|S_{c,q}| \neq 0$ from Lemma 1. We will show that $\psi_{c,q}$ satisfies the following three conditions:

(Normalization) For the multiplicative identity $\mathbf{1} \in \text{Pol}_2(\mathbb{F}_q)$, we have $\psi_{c,q}(\mathbf{1}) = 1_q$.

(Locality) If π is a polynumber for which $\varepsilon(\pi)$ is the zero function when restricted to $S_{c,q}$, then $\psi_{c,q}(\pi) = 0$.

(Invariance) $\psi_{c,q}$ is rotationally invariant with respect to the group of linear isometries of the affine space determined by the geometry.

Any linear functional satisfying the three conditions above will be called a **circular integral functional**. A main result is the following uniqueness theorem.

Theorem 1. *The Fourier summation functional $\psi_{c,q}$ is the unique circular integral functional in the geometry with color c , for each of the green, red, and blue geometries.*

The second half of this paper is dedicated to finding an explicit formula for $\psi_{c,q}$ in each case. The result in the blue (Euclidean) geometry is established from those in the green and red (relativistic) geometries. The explicit formulas for $\psi_{c,q}$ in the red and blue geometries involve the circular super Catalan numbers $\Omega(m, n)$ reduced modulo p , which we think of as elements of $\mathbb{F}_p$. Note that the expression

$$\Omega(m, n) \bmod p = \frac{S(m, n)}{4^{m+n}} \bmod p$$

is well-defined since $S(m, n)$ is a natural number and 4^{m+n} is never a multiple of p . A simplified and restricted version of the formulas for $\psi_{c,q}$ is given below.

Theorem 2. *Consider a finite field $\mathbb{F}_q$ where $q = p^r$ for some prime $p > 2$.*

1. *In the red geometry, for $0 \leq k + l < q - 1$,*

$$\psi_{r,q}(\alpha^k \beta^l) = \begin{cases} (-1)^n \Omega(m, n) \bmod p & \text{if } k = 2m \text{ and } l = 2n, \\ 0 & \text{otherwise.} \end{cases}$$

2. *In the blue geometry, for $0 \leq k + l < q - 1$,*

$$\psi_{b,q}(\alpha^k \beta^l) = \begin{cases} \Omega(m, n) \bmod p & \text{if } k = 2m \text{ and } l = 2n, \\ 0 & \text{otherwise.} \end{cases}$$

As a consequence, the normalized sums $\psi_{r,q}(\alpha^k \beta^l)$ and $\psi_{b,q}(\alpha^k \beta^l)$ lie in the prime field $\mathbb{F}_p$, and are independent of the power q of p . Moreover, $\psi_{r,q}(\alpha^{2m} \beta^{2n})$ and $\psi_{b,q}(\alpha^{2m} \beta^{2n})$ are never 0. Perhaps most remarkably, the single rational number $\Omega(m, n)$ determines the values $\psi_{r,q}(\alpha^{2m} \beta^{2n})$ and $\psi_{b,q}(\alpha^{2m} \beta^{2n})$ for *all* prime powers $q > 2(m + n) + 1$.

Here is an example to illustrate these formulas.

Example 1. We work with $\mathbb{F}_{13}$ which we identify as $\{0, 1, 2, \dots, 12\}$. Here, $-1 = 12$ is a square so by Lemma 1, $|S_{b,13}| = 12 = -1$ regarded as an element of $\mathbb{F}_{13}$. Using Theorem 2, if $0 \leq m + n < 6$ then $\psi_{b,13}(\alpha^{2m} \beta^{2n}) = \Omega(m, n) \bmod 13$. The blue unit circle is

$$S_{b,13} = \{[1, 0] \ [12, 0] \ [0, 1] \ [0, 12] \ [2, 6] \ [2, 7] \ [11, 6] \ [11, 7] \ [6, 2] \ [7, 2] \ [6, 11] \ [7, 11]\}$$

where there is no need for commas as the points are not naturally ordered. By manual inspection, and omitting the zero terms,

$$\begin{aligned}\psi_{b,13}(\alpha^2\beta^6) &= - \sum_{[x,y] \in S_{b,13}} x^2 y^6 \\ &= - (2^2 6^6 + 2^2 7^6 + 11^2 6^6 + 11^2 7^6 + 6^2 2^6 + 7^2 2^6 + 6^2 11^6 + 7^2 11^6) = 4.\end{aligned}$$

We may verify that $\Omega(1,3) \bmod 13 = \frac{5}{128} \bmod 13 = 4$.

The complete formulas for the red and blue geometries which go beyond the constraint $k+l < q-1$ are more involved, and they point towards an additional family of integers that extend the super Catalan numbers, and that arise as coefficients of what we call the **circular polynumbers**

$$\pi_{k,l} \equiv \left(\frac{1+\alpha}{2} \right)^k \left(\frac{1-\alpha}{2} \right)^l$$

suitably reduced modulo p . The coefficient of α^n of $\pi_{k,l}$ is denoted by $[\alpha^n] \pi_{k,l}$ and for a general polynumber π we also write

$$\left(\sum_{i=1}^l [\alpha^{k_i}] \right) \pi \equiv [\alpha^{k_1}] \pi + [\alpha^{k_2}] \pi + \cdots + [\alpha^{k_l}] \pi.$$

The key point that connects these circular polynumbers and the super Catalan numbers is the following restatement of the main result of Georgiadis, Munemasa, and Tanaka in [14] in terms of $\Omega(m,n)$ and $\pi_{k,l}$.

Theorem 3. *For any natural numbers m and n ,*

$$\Omega(m,n) = (-1)^n [\alpha^{m+n}] \pi_{2m,2n}.$$

The complete summation formulas for $\psi_{c,q}$ are presented separately as Theorems 12, 13, and 17, which we summarize below. They involve geometric constants that depend on the sizes of the unit circles in each geometry. As usual, $\lfloor x \rfloor$ denotes the usual floor function.

Theorem 4. *Consider the finite field $\mathbb{F}_q$ of size $q = p^r$ with $p > 2$.*

1. *In the green geometry,*

$$\psi_{g,q}(\alpha^k \beta^l) = \begin{cases} 1_q & \text{if } |S_{g,q}| \mid (k-l), \\ 0 & \text{otherwise.} \end{cases}$$

2. *In the red geometry, with $w \equiv \frac{1}{2} |S_{r,q}|$ and $R \equiv \left\lfloor \frac{k+l}{|S_{r,q}|} \right\rfloor$,*

$$\psi_{r,q}(\alpha^k \beta^l) = \begin{cases} \left(\sum_{|d| \leq R} [\alpha^{m+n+dw}] \right) \pi_{k,l} \bmod p & \text{if } k = 2m, l = 2n, \\ 0 & \text{otherwise.} \end{cases}$$

3. In the blue geometry, with $w \equiv \frac{1}{2} |S_{b,q}|$ and $R \equiv \left\lfloor \frac{k+l}{|S_{b,q}|} \right\rfloor$,

$$\psi_{b,q}(\alpha^k \beta^l) = \begin{cases} \left(\sum_{|d| \leq R} [\alpha^{m+n+dw}] \right) (-1)^n \pi_{k,l} \bmod p & \text{if } k = 2m, l = 2n, \\ 0 & \text{otherwise.} \end{cases}$$

Thus we see that the complete formulas to evaluate $\psi_{r,q}(\alpha^k \beta^l)$ and $\psi_{b,q}(\alpha^k \beta^l)$ involve an equally-spaced ladder of coefficients of $\pm \pi_{k,l}$ symmetrically placed around the central coefficient which is, by the result of Georgiadis, Munemasa, and Tanaka, $\Omega(m, n)$ when $k = 2m$ and $l = 2n$.

To illustrate, we will compute the value of $\psi_{b,q}$ evaluated at $\alpha^6 \beta^2$ over two different fields $\mathbb{F}_{13}$ and $\mathbb{F}_7$. Both computations involve certain coefficients of the polynumber

$$\begin{aligned} -\pi_{6,2} &= - \left(\frac{1+\alpha}{2} \right)^6 \left(\frac{1-\alpha}{2} \right)^2 \\ &= -\frac{1}{256} \alpha^8 - \frac{1}{64} \alpha^7 - \frac{1}{64} \alpha^6 + \frac{1}{64} \alpha^5 + \frac{5}{128} \alpha^4 + \frac{1}{64} \alpha^3 - \frac{1}{64} \alpha^2 - \frac{1}{64} \alpha - \frac{1}{256}. \end{aligned}$$

Note that this expression is symmetric around the central term of α^4 , where the coefficient is $\frac{5}{128} = \Omega(3, 1)$.

Example 2. Continuing our previous example where $q = 13$, we deduce that $w = \frac{1}{2} |S_{b,13}| = 6$ and $R_{k,l} = \left\lfloor \frac{k+l}{|S_{b,13}|} \right\rfloor = 0$. The formula for $\psi_{b,q}$ in Theorem 4 above then reads

$$\psi_{b,13}(\alpha^6 \beta^2) = \left(\sum_{|d| \leq 0} [\alpha^{4+6d}] \right) (-\pi_{8,2}) \bmod 13 = [\alpha^4] (-\pi_{8,2}) \bmod 13 = 4.$$

Here we see that the condition $0 \leq k+l < q-1$ reduces the sum to just the central term.

However when $q = 7$, -1 is not a square so by Lemma 1, $w = \frac{1}{2} |S_{b,7}| = 4$ and $R_{k,l} = \left\lfloor \frac{k+l}{|S_{b,7}|} \right\rfloor = 1$. It follows that

$$\begin{aligned} \psi_{b,7}(\alpha^6 \beta^2) &= \left(\sum_{|d| \leq 1} [\alpha^{4+4d}] \right) (-\pi_{8,2}) \bmod 7 \\ &= ([\alpha^0] + [\alpha^4] + [\alpha^8]) (-\pi_{8,2}) \bmod 7 \\ &= \left(-\frac{1}{256} + \frac{5}{128} - \frac{1}{256} \right) \bmod 7 = \frac{1}{32} \bmod 7 = 2. \end{aligned}$$

It can be verified that $S_{b,7} = \{[1, 0] \ [6, 0] \ [0, 1] \ [0, 1] \ [2, 2] \ [5, 5] \ [2, 5] \ [5, 2]\}$ and by manual computation $\psi_{b,7}(\alpha^6 \beta^2) = 2^6 2^2 + 5^6 5^2 + 2^6 5^2 + 5^6 2^2 = 2$.

1.2 Connection to characteristic zero integration theory

A formula for the integral of a polynomial over a general sphere was given by Baker [2] and later by Folland [13], both of which involve the gamma function Γ . Here we restate the main theorem in Folland's paper involving the usual rotationally-invariant measure $d\mu$ on the $(d-1)$ -dimensional unit sphere in $\mathbb{R}^d$ coming from the standard Riemannian structure on $\mathbb{R}^d$, without proof.

Theorem 5. *Let $S_d = \{[x_1, x_2, \dots, x_d] \in \mathbb{R}^d : x_1^2 + x_2^2 + \dots + x_d^2 = 1\}$. We have that*

$$\int_{S_d} x_1^{a_1} x_2^{a_2} \dots x_d^{a_d} d\mu = \begin{cases} \frac{2}{\Gamma(b_1 + b_2 + \dots + b_d)} \prod_{i=1}^d \Gamma(b_i) & \text{if } (a_1, a_2, \dots, a_n) = (2b_1, 2b_2, \dots, 2b_n) \\ 0 & \text{otherwise.} \end{cases}$$

Importantly, we note that the formula in Theorem 5 above can be reformulated without reference to Γ and π , thereby eliminating the transcendental aspect, but the key is to renormalize the integral. We illustrate this in the specific case $d = 2$ involving the usual (blue) unit circle. Indeed, by noting that $\Gamma(x+1) = x\Gamma(x)$ and $\Gamma(\frac{1}{2}) = \sqrt{\pi}$, we obtain a formula for the integral when $(a_1, a_2) = (2m, 2n)$ as

$$\begin{aligned} \frac{2\Gamma(b_1)\Gamma(b_2)}{\Gamma(b_1+b_2)} &= \frac{2\Gamma(m+\frac{1}{2})\Gamma(n+\frac{1}{2})}{\Gamma(m+n+1)} \\ &= \frac{2}{(m+n)!} \left(\prod_{k=1}^m \left(\frac{2k-1}{2} \right) \right) \left(\prod_{l=1}^n \left(\frac{2l-1}{2} \right) \right) \Gamma\left(\frac{1}{2}\right)^2 \\ &= \frac{2\pi}{2^{m+n}} \frac{(2m)!}{(m+n)!} \frac{(2n)!}{2^m m! 2^n n!} \\ &= 2\pi\Omega(m, n). \end{aligned}$$

If the integral over the unit circle S_2 is normalized, then

$$\int_{S_2} x_1^{a_1} x_2^{a_2} d\mu = \begin{cases} \Omega(m, n) & \text{if } a_1 = 2m, a_2 = 2n, \\ 0 & \text{otherwise} \end{cases}$$

which is strikingly similar to the formula for $\psi_{b,q}$ presented in Theorem 2.

2 Polynomials and functions over finite fields

Since we are dealing here with finite mathematics and finite arithmetic, we propose to dispense with the usual reliance on "infinite sets". Throughout this paper, $\mathbb{N}$ will denote the *type* of a natural number — which for us includes the number 0; $\mathbb{Z}$ denotes the type of integers; $\mathbb{Q}$ denotes the type of rational numbers; and $\mathbb{F}_q$ will denote the finite field of size q , where q is a power of an odd prime p . We may express the multiplicative identity in $\mathbb{F}_q$ as 1_q if there is possibility of confusion. The notation $\mathbb{F}^\times$ is reserved for the multiplicative group of the field $\mathbb{F}$. We also use the convention that $0^0 = 1$.

We briefly distinguish between polynomials as formal expressions and as functions by regarding the former purely as finite lists of numbers in $\mathbb{F}$, thereby avoiding the notion of "variables". So a

polynomial π over a field $\mathbb{F}$ is a finite array of numbers in $\mathbb{F}$ written vertically with a bar above it and another one to the left, as in

$$\pi = \overline{\begin{array}{c} u_0 \\ u_1 \\ \vdots \\ u_k \end{array}}.$$

Adding or removing zeroes from the bottom will not change a polynomial. The **zero polynomial** and the **identity polynomial** are respectively $\mathbf{0} \equiv \overline{0}$ and $\mathbf{1} \equiv \overline{1}$. The addition and scalar multiplication of polynomials are entry-wise while polynomial multiplication is defined as the usual Cauchy product on sequence of numbers. The notion of degree and irreducibility of a polynomial is similar to that of a polynomial, with the degree of $\mathbf{0}$ not defined. We denote the algebra of polynomials over $\mathbb{F}$ by $\text{Pol}(\mathbb{F})$.

After we define the special polynomial

$$\alpha \equiv \overline{\begin{array}{c} 0 \\ 1 \end{array}}$$

we say a **polynomial** in α is a polynomial written as a linear combination of powers of the polynomial α . If $\pi = u_0\mathbf{1} + u_1\alpha + \cdots + u_k\alpha^k$ is a polynomial in α , then u_k is the coefficient of α^k in π , and we write $u_k = [\alpha^k] \pi$. We will allow ourselves to use also summation notation and write

$$\left(\sum_{i=1}^l [\alpha^{k_i}] \right) \pi \equiv [\alpha^{k_1}] \pi + \cdots + [\alpha^{k_l}] \pi.$$

In a similar fashion, we can define a **2-polynomial** over $\mathbb{F}$ to be a two-dimensional array of numbers in $\mathbb{F}$ of the form

$$\pi \equiv \overline{\begin{array}{cccc} u_{00} & u_{01} & \cdots & u_{0n} \\ u_{10} & u_{11} & \cdots & u_{1n} \\ \vdots & \vdots & \ddots & \vdots \\ u_{m0} & u_{m1} & \cdots & u_{mn} \end{array}}$$

with the convention that adding further zeroes to the right and down leaves the 2-polynomial unchanged. The algebra operations are analogous to those in $\text{Pol}(\mathbb{F})$, with the multiplication defined by the two-dimensional version of the Cauchy product. The algebra of 2-polynomials over $\mathbb{F}$ is denoted by $\text{Pol}_2(\mathbb{F})$. The multiplicative identity is also denoted by $\mathbf{1}$, not to be confused with that of $\text{Pol}(\mathbb{F})$ if the context is clear. After we define the polynomial

$$\beta \equiv \overline{\begin{array}{cc} 0 & 1 \end{array}}$$

any 2-polynomial can be written as a linear combination of powers of α and β and then the algebra operations agree with the familiar operations on polynomials. Note however that here and throughout

α and β are *not* variables; they are just particular polynumbers. Clearly we may regard $\text{Pol}(\mathbb{F})$ as a subalgebra of $\text{Pol}_2(\mathbb{F})$.

If $\pi \in \text{Pol}(\mathbb{F}_p)$ is an irreducible polynumber of degree r and (π) denotes the ideal of $\text{Pol}(\mathbb{F}_p)$ generated by π , then $\text{Pol}(\mathbb{F}_p)/(\pi)$ is a field of size $q = p^r$. Since any two finite fields of the same size are isomorphic, the choice of π does not matter.

The two-dimensional **affine plane** is denoted by $\mathbb{A} = \mathbb{A}(\mathbb{F}) = \{[x, y] : x, y \in \mathbb{F}\}$, with the objects $[a, b]$ called **affine points**, or just **points**. Moreover, the space of functions from $\mathbb{A}$ to $\mathbb{F}$, denoted by $\mathbb{F}^{\mathbb{A}}$, is an algebra over $\mathbb{F}$ under the usual pointwise operation. For brevity, we write $f([x, y])$ simply as $f(x, y)$.

If $\mathbb{F} = \mathbb{F}_q$ is a finite field with $q = p^r$ elements, then $\dim(\mathbb{F}_q^{\mathbb{A}}) = q^2$ and a basis for $\mathbb{F}_q^{\mathbb{A}}$ is $\Delta = \{\delta_{[a, b]} : [a, b] \in \mathbb{A}\}$, where

$$\delta_{[a, b]}(x, y) = \begin{cases} 1_q & \text{if } [x, y] = [a, b] \\ 0 & \text{if } [x, y] \neq [a, b]. \end{cases}$$

The multiplicative identity in $\mathbb{F}_q^{\mathbb{A}}$ is $1_{\mathbb{F}^{\mathbb{A}}}$. Since $\delta_{[a, b]}(x, y)$ vanishes at every point in $\mathbb{A}$ except $[a, b]$, we have

$$1_{\mathbb{F}^{\mathbb{A}}} = \sum_{[a, b] \in \mathbb{A}} \delta_{[a, b]}. \quad (1)$$

Evaluation of a 2-polynumber in $\text{Pol}_2(\mathbb{F}_q)$ then gives a linear map $\varepsilon : \text{Pol}_2(\mathbb{F}_q) \rightarrow \mathbb{F}_q^{\mathbb{A}}$. Although clearly not injective, this map is onto. To see this, we set for any $[a, b] \in \mathbb{A}$ the 2-polynumber

$$\pi_{[a, b]} \equiv \left(\prod_{\substack{t \in \mathbb{F}_q \\ t \neq a}} (t - a) \prod_{\substack{u \in \mathbb{F}_q \\ u \neq b}} (u - b) \right)^{-1} \prod_{\substack{t \in \mathbb{F}_q \\ t \neq a}} (t - \alpha) \prod_{\substack{u \in \mathbb{F}_q \\ u \neq b}} (u - \beta)$$

and observe that $\varepsilon(\pi_{[a, b]}) = \delta_{[a, b]}$.

The subspace $\mathcal{P}_q = \text{span} \{\alpha^k \beta^l : 0 \leq k, l \leq q-1\}$ of $\text{Pol}_2(\mathbb{F}_q)$ has dimension q^2 and we call it the **principal subspace**. Note that for any point $[a, b]$ we have $\pi_{[a, b]} \in \mathcal{P}_q$. Since $\dim(\mathcal{P}_q) = \dim(\mathbb{F}_q^{\mathbb{A}})$ and $\varepsilon|_{\mathcal{P}_q}$ is onto, it is a vector space isomorphism. In particular, the set $\Pi = \{\pi_{[a, b]} : [a, b] \in \mathbb{A}\}$ is a basis for $\mathcal{P}_q$. The following result gives a vector space decomposition of $\text{Pol}_2(\mathbb{F}_q)$.

Proposition 1. *We have that $\text{Pol}_2(\mathbb{F}_q) = \mathcal{P}_q \oplus \ker(\varepsilon)$.*

Proof. Take any $\pi \in \text{Pol}_2(\mathbb{F}_q)$. Since $\varepsilon|_{\mathcal{P}_q}$ is an isomorphism, there is a unique $\pi_1 = (\varepsilon|_{\mathcal{P}_q}^{-1} \circ \varepsilon)(\pi) \in \mathcal{P}_q$ for which $\varepsilon(\pi_1) = \varepsilon(\pi)$. By linearity, $\pi - \pi_1 \in \ker(\varepsilon)$ and therefore the decomposition $\pi_1 + (\pi - \pi_1)$ is unique. $\square$

2.1 Group actions on various spaces

We now define some group actions on the relevant spaces that will be useful later in our analysis. The traditional treatment to specifying a group action is by fixing a direction the group acts from, that is, agreeing that every action will be a left action or a right action. We will instead be more flexible and specify explicitly whether a given action is a left action or a right action.

We say that a group G with identity e **left acts** on a space $\mathbb{X}$ precisely when for any $g \in G$ and $X \in \mathbb{X}$ there is an element $g \cdot X \in \mathbb{X}$ such that $e \cdot X = X$ and $h \cdot (g \cdot X) = (hg) \cdot X$ for $X \in \mathbb{X}$ and $g, h \in G$. We say that a group G with identity e **right acts** on a space $\mathbb{X}$ precisely when for any $g \in G$ and $X \in \mathbb{X}$ there is an element $X \cdot g \in \mathbb{X}$ such that $X \cdot e = X$ and $(X \cdot g) \cdot h = X \cdot (gh)$ for $X \in \mathbb{X}$ and $g, h \in G$.

From our point of view, the distinction between these two types of actions has at least two benefits. First, notions of duality and functoriality can generally be more easily appreciated. Secondly, this approach generally avoids the need for inverses, which is convenient since they are generally computationally expensive and thus should be avoided theoretically whenever possible.

Now suppose that G is a group with identity e that right acts on the affine plane $\mathbb{A}$. Then there is a natural left action of G on the space of functions $\mathbb{F}_q^\mathbb{A}$ defined for $h \in G$ and $f \in \mathbb{F}_q^\mathbb{A}$ by $(h \cdot f)(x, y) \equiv f([x, y] \cdot h)$. We can quickly verify that the basis Δ is invariant under this left action: $(h \cdot \delta_{[a, b]})(x, y) = \delta_{[a, b]}([x, y] \cdot h)$ for any $h \in G$ and $[a, b] \in \mathbb{A}$, and therefore

$$h \cdot \delta_{[a, b]} = \delta_{[a, b] \cdot h^{-1}}. \quad (2)$$

We say that $f \in \mathbb{F}_q^\mathbb{A}$ is **G -invariant** precisely when $h \cdot f = f$ for any $h \in G$, and let $\mathcal{I}_G$ be the subalgebra of $\mathbb{F}_q^\mathbb{A}$ consisting of G -invariant functions. Note that for any $f \in \mathbb{F}_q^\mathbb{A}$, the **G -average** of f , defined by

$$\rho_G(f) \equiv \frac{1}{|G|} \sum_{h \in G} (h \cdot f)$$

is always G -invariant, provided that $|G| \neq 0$ in $\mathbb{F}_q$. Indeed, for any $g \in G$ and $[x, y] \in \mathbb{A}$, we have

$$\begin{aligned} (g \cdot \rho_G(f))(x, y) &= \rho_G(f)([x, y] \cdot g) \\ &= \frac{1}{|G|} \sum_{h \in G} (h \cdot f)([x, y] \cdot g) = \frac{1}{|G|} \sum_{h \in G} (gh \cdot f)(x, y) \\ &= \frac{1}{|G|} \sum_{h \in G} (h \cdot f)(x, y) \quad (\text{via a change of index } h \mapsto g^{-1}h) \\ &= \rho_G(f)(x, y). \end{aligned}$$

Moreover, it is obvious that $\rho_G(f) = f$ if $f \in \mathcal{I}_G$. This makes ρ_G a projection onto $\mathcal{I}_G$.

Now there is also a natural right action of G on the dual space $(\mathbb{F}_q^\mathbb{A})^*$ of $\mathbb{F}_q^\mathbb{A}$ by $(\phi \cdot h)(f) \equiv \phi(h \cdot f)$ for any $\phi \in (\mathbb{F}_q^\mathbb{A})^*$, $h \in G$, and $f \in \mathbb{F}_q^\mathbb{A}$. Here the duality between $\mathbb{F}_q^\mathbb{A}$ and $(\mathbb{F}_q^\mathbb{A})^*$ is also reflected in the left and right action of G on $\mathbb{F}_q^\mathbb{A}$ and $(\mathbb{F}_q^\mathbb{A})^*$.

A linear functional $\phi \in (\mathbb{F}_q^\mathbb{A})^*$ is **G -invariant** precisely when $\phi \cdot h = \phi$ for any $h \in G$. The subspace of G -invariant functionals in $(\mathbb{F}_q^\mathbb{A})^*$ can then be identified with $\mathcal{I}_G^*$. By a similar argument, the **G -average** of ϕ defined by

$$\hat{\rho}_G(\phi) \equiv \frac{1}{|G|} \sum_{h \in G} (\phi \cdot h)$$

provided that $|G| \neq 0$ in $\mathbb{F}_q$ is **G -invariant** and therefore $\hat{\rho}_G$ is a projection of $(\mathbb{F}_q^\mathbb{A})^*$ onto $\mathcal{I}_G^*$.

2.2 Krawtchouk polynomials and the circular polynomials

For $l \in \mathbb{N}$, we define the rational polynomial

$$\binom{\alpha}{l} \equiv \begin{cases} \frac{\alpha(\alpha-1)(\alpha-2)\dots(\alpha-l+1)}{l!} & \text{if } l \geq 1 \\ 1 & \text{if } l = 0. \end{cases}$$

With this notation, the composition of $\binom{\alpha}{l}$ for $l \geq 1$ with another rational polynomial π can be written as

$$\binom{\pi}{l} = \frac{\pi(\pi-1)(\pi-2)\dots(\pi-l+1)}{l!}.$$

For $0 \leq n \leq d$, the n -th **Krawtchouk polynomial** $k_n^{(d)}$ of order d is given by

$$k_n^{(d)}(\alpha) \equiv \sum_{l=0}^n (-1)^l \binom{\alpha}{l} \binom{d-\alpha}{n-l}.$$

This family of polynomials generalizes the Hermite polynomials and was introduced by Mikhail Krawtchouk in the late 1920s [19]. Their generating function is particularly pleasant, given by

$$\sum_{n=0}^d k_n^{(d)}(m) \alpha^n = (1-\alpha)^m (1+\alpha)^{d-m} \quad (3)$$

where $k_n^{(d)}(m)$ is the evaluation of $k_n^{(d)}$ at $m \in \{0, 1, 2, \dots, d\}$.

Krawtchouk polynomials play an important role in many areas of mathematics, such as harmonic analysis in the context of orthogonal polynomials [10], [11], coding theory in the context of bounds of codes [20], [21], [23], and probability theory in the context of symmetric random walk [12].

In 2011, Georgiadis, Munemasa, and Tanaka in their paper [14] gave an interpretation of $S(m, n)$ in terms of values of Krawtchouk polynomials. We state their main result now, and will provide a new proof of it shortly.

Theorem 6 (Georgiadis, Munemasa, Tanaka). *For any natural numbers m and n ,*

$$k_{m+n}^{(2m+2n)}(2m) = (-1)^m S(m, n). \quad (4)$$

The next proposition is a restatement of Theorem 6 in terms of the generating function.

Proposition 2. *For any natural numbers m and n ,*

$$S(m, n) = (-1)^n [\alpha^{m+n}] (1+\alpha)^{2m} (1-\alpha)^{2n}.$$

Proof. From (3) and (4), we deduce that

$$(-1)^m S(m, n) = [\alpha^{m+n}] (1-\alpha)^{2m} (1+\alpha)^{2n}. \quad (5)$$

Finally, by multiplying both sides of equation (5) above by $(-1)^m$, we obtain

$$\begin{aligned} S(m, n) &= (-1)^m [\alpha^{m+n}] (1-\alpha)^{2m} (1+\alpha)^{2n} \\ &= (-1)^n [(-\alpha)^{m+n}] (1-\alpha)^{2m} (1+\alpha)^{2n} \\ &= [\alpha^{m+n}] (1+\alpha)^{2m} (1-\alpha)^{2n}. \end{aligned}$$

□

We now introduce a central family of polynomials called the **circular polynomials**

$$\pi_{k,l} \equiv \left(\frac{1+\alpha}{2} \right)^k \left(\frac{1-\alpha}{2} \right)^l$$

for $k, l \in \mathbb{N}$. These polynomials, as we shall soon see, are crucial in describing the Fourier summation functional formulas in the red and blue geometries.

It is clear that $\pi_{0,0} = \mathbf{1}$. Here is a table of the polynomials $\pi_{k,l}$ for the next few values of k and l , organized by the sum $k+l$:

$\pi_{k,l}$	$k+l=1$	$\pi_{k,l}$	$k+l=2$	$\pi_{k,l}$	$k+l=3$
$\pi_{0,1}$	$\frac{1}{2}(1-\alpha)$	$\pi_{0,2}$	$\frac{1}{4}(1-2\alpha+\alpha^2)$	$\pi_{0,3}$	$\frac{1}{8}(1-3\alpha+3\alpha^2-\alpha^3)$
$\pi_{1,0}$	$\frac{1}{2}(1+\alpha)$	$\pi_{1,1}$	$\frac{1}{4}(1-\alpha^2)$	$\pi_{1,2}$	$\frac{1}{8}(1-\alpha-\alpha^2+\alpha^3)$
		$\pi_{2,0}$	$\frac{1}{4}(1+2\alpha+\alpha^2)$	$\pi_{2,1}$	$\frac{1}{8}(1+\alpha-\alpha^2-\alpha^3)$
				$\pi_{3,0}$	$\frac{1}{8}(1+3\alpha+3\alpha^2+\alpha^3)$

$\pi_{k,l}$	$k+l=4$	$\pi_{k,l}$	$k+l=5$
$\pi_{0,4}$	$\frac{1}{16}(1-4\alpha+6\alpha^2-4\alpha^3+\alpha^4)$	$\pi_{0,5}$	$\frac{1}{32}(1-5\alpha+10\alpha^2-10\alpha^3+5\alpha^4-\alpha^5)$
$\pi_{1,3}$	$\frac{1}{16}(1-2\alpha+2\alpha^3-\alpha^4)$	$\pi_{1,4}$	$\frac{1}{32}(1-3\alpha+2\alpha^2+2\alpha^3-3\alpha^4+\alpha^5)$
$\pi_{2,2}$	$\frac{1}{16}(1-2\alpha^2+\alpha^4)$	$\pi_{2,3}$	$\frac{1}{32}(1-\alpha-2\alpha^2+2\alpha^3+\alpha^4-\alpha^5)$
$\pi_{3,1}$	$\frac{1}{16}(1+2\alpha-2\alpha^3-\alpha^4)$	$\pi_{3,2}$	$\frac{1}{32}(1+\alpha-2\alpha^2-2\alpha^3+\alpha^4+\alpha^5)$
$\pi_{4,0}$	$\frac{1}{16}(1+4\alpha+6\alpha^2+4\alpha^3+\alpha^4)$	$\pi_{4,1}$	$\frac{1}{32}(1+3\alpha+2\alpha^2-2\alpha^3-3\alpha^4-\alpha^5)$
		$\pi_{5,0}$	$\frac{1}{32}(1+5\alpha+10\alpha^2+10\alpha^3+5\alpha^4+\alpha^5)$

Table 5: The circular polynomials $\pi_{k,l}$ for $0 \leq k+l \leq 5$.

We say that a polynomial $\pi = u_0\alpha^0 + u_1\alpha^1 + \dots + u_{d-1}\alpha^{d-1} + u_d\alpha^d$ of degree d is **palindromic** precisely when $u_k = u_{d-k}$ and **anti-palindromic** precisely when $u_k = -u_{d-k}$, for all $0 \leq k \leq d$. The product of two palindromic polynomials or two anti-palindromic polynomials is palindromic, while the product of a palindromic and an anti-palindromic polynomial is anti-palindromic.

Lemma 2 (Palindromic and symmetry properties of $\pi_{k,l}$). *If $0 \leq i, j \leq k+l$ and $i+j = k+l$ then*

$$[\alpha^i] \pi_{k,l} = (-1)^l [\alpha^j] \pi_{k,l}.$$

Moreover, if $0 \leq i \leq k+l$ then

$$[\alpha^i] \pi_{k,l} = (-1)^i [\alpha^i] \pi_{l,k}.$$

Proof. Clearly $\pi_{k,0}$ is a palindromic polynomial for any k , while $\pi_{0,l}$ is palindromic if l is even and anti-palindromic if l is odd. The first identity follows from $\pi_{k,l} = \pi_{k,0}\pi_{0,l}$. For the second identity, we observe that $\pi_{l,k}(-\alpha) = \pi_{k,l}(\alpha)$. Thus by extracting the coefficient of α^i from both sides,

$$[\alpha^i] \pi_{k,l} = [\alpha^i] \pi_{l,k}(-\alpha) = [(-\alpha)^i] \pi_{l,k} = [(-1)^i \alpha^i] \pi_{l,k} = (-1)^i [\alpha^i] \pi_{l,k}. \quad \square$$

We now restate Theorem 6 in terms of the circular super Catalan numbers $\Omega(m, n)$ and the central coefficient of the circular polynumbers $\pi_{2m, 2n}$, and give an independent algebraic proof. This allows us to naturally identify the remaining coefficients around the central one as secondary but still important variants of the circular super Catalan numbers.

Theorem 7. *For any natural numbers m and n ,*

$$\Omega(m, n) = (-1)^n [\alpha^{m+n}] \pi_{2m, 2n}.$$

Proof. We calculate that

$$\begin{aligned} (-1)^n [\alpha^{m+n}] \pi_{2m, 2n} &= (-1)^n [\alpha^{m+n}] \left(\left(\frac{1+\alpha}{2} \right)^{2m} \left(\frac{1-\alpha}{2} \right)^{2n} \right) \\ &= \frac{(-1)^n}{4^{m+n}} \sum_{\substack{0 \leq s \leq 2m \\ 0 \leq t \leq 2n \\ s+t=m+n}} (-1)^t \binom{2m}{s} \binom{2n}{t}. \end{aligned}$$

By using the convention that $\binom{k}{s} = 0$ if $s < 0$ or $s > k$, we may rearrange this sum as

$$\begin{aligned} (-1)^n [\alpha^{m+n}] \pi_{2m, 2n} &= \frac{(-1)^n}{4^{m+n}} \sum_{s=0}^{m+n} (-1)^{m+n-s} \binom{2m}{s} \binom{2n}{m+n-s} \\ &= \frac{(-1)^m}{4^{m+n}} \sum_{s=0}^{m+n} (-1)^s \frac{(2m)!}{s! (2m-s)!} \frac{(2n)!}{(m+n-s)! (-m+n+s)!} \\ &= \frac{(-1)^m (2m)! (2n)!}{4^{m+n} (m+n)! (m+n)!} \sum_{s=0}^{m+n} (-1)^s \binom{m+n}{s} \binom{m+n}{2m-s}. \end{aligned} \quad (6)$$

The summation on the right-hand side of (6) can be realized as

$$[\alpha^{2m}] (1+\alpha)^{m+n} (1-\alpha)^{m+n} = [\alpha^{2m}] (1-\alpha^2)^{m+n}$$

so we get

$$\begin{aligned} (-1)^n [\alpha^{m+n}] \pi_{2m, 2n} &= \frac{(-1)^m (2m)! (2n)!}{4^{m+n} (m+n)! (m+n)!} [\alpha^{2m}] (1-\alpha^2)^{m+n} \\ &= \frac{(2m)! (2n)!}{4^{m+n} (m+n)! (m+n)!} \binom{m+n}{m} = \Omega(m, n). \end{aligned} \quad \square$$

3 Chromogeometry and Dihedrons

We use chromogeometry, a three-fold symmetry introduced by Wildberger in [37], as a foundation for Fourier summation over finite fields, involving both the unit circles over which Fourier summation takes place, and the associated groups of rotation matrices which act on the space of 2-polynumbers. The three geometries will be defined through symmetric bilinear forms induced by symmetric matrices, thereby avoiding notions of distance involving square roots, which do not generally apply to finite fields.

If $\mathbb{F}$ is a field of characteristic not two, then the algebra of 2×2 matrices over $\mathbb{F}$, which we call the **dihedron algebra** over $\mathbb{F}$, is denoted by $\mathbb{D} = \mathbb{D}(\mathbb{F})$. This is a four-dimensional algebra over $\mathbb{F}$ with a special basis consisting of the four matrices

$$1_{\mathbb{D}} \equiv \begin{pmatrix} 1_{\mathbb{F}} & 0 \\ 0 & 1_{\mathbb{F}} \end{pmatrix} \quad i \equiv \begin{pmatrix} 0 & 1_{\mathbb{F}} \\ -1_{\mathbb{F}} & 0 \end{pmatrix} \quad j \equiv \begin{pmatrix} 0 & 1_{\mathbb{F}} \\ 1_{\mathbb{F}} & 0 \end{pmatrix} \quad k \equiv \begin{pmatrix} 1_{\mathbb{F}} & 0 \\ 0 & -1_{\mathbb{F}} \end{pmatrix}.$$

The dihedron algebra terminology is motivated by the observation that $\{\pm 1_{\mathbb{D}}, \pm i, \pm j, \pm k\}$ is the dihedral group D_4 , and was coined by Wildberger in his online *Linear Algebra* lectures series. The situation is analogous to and shares many parallels with the quaternion group Q also of order 8 and the associated four-dimensional algebra $\mathbb{H}$ of quaternions. We can think of $\mathbb{F}$ as being embedded in $\mathbb{D}$ by making the identification $t \mapsto t1_{\mathbb{D}}$.

These four special basis matrices satisfy

$$\begin{aligned} i^2 &= -1_{\mathbb{D}} & j^2 &= 1_{\mathbb{D}} & k^2 &= 1_{\mathbb{D}} \\ jk &= -i & ki &= j & ij &= k \\ kj &= i & ik &= -j & ji &= -k. \end{aligned}$$

The difference between $\mathbb{D}$ and $\mathbb{H}$ then amounts to small alterations in signs for the structure constants when comparing the basis elements $1_{\mathbb{D}}, i, j$, and k and the more usual corresponding elements in Q . We equip $\mathbb{D}$ with the structure of a ***-algebra** over $\mathbb{F}$ by defining the involution to be the **adjugate** map

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix}^* \equiv \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$$

and note that for $h_1, h_2 \in \mathbb{D}$ we have $(h_1 h_2)^* = h_2^* h_1^*$. It is also obvious that $h h^* = h^* h = \det(h) 1_{\mathbb{D}}$ for all $h \in \mathbb{D}$.

There are three particularly useful commutative *-subalgebras of $\mathbb{D}(\mathbb{F})$, namely

$$\mathbb{C}_b(\mathbb{F}) \equiv \text{span}\{1_{\mathbb{D}}, i\} \quad \mathbb{C}_r(\mathbb{F}) \equiv \text{span}\{1_{\mathbb{D}}, j\} \quad \mathbb{C}_g(\mathbb{F}) \equiv \text{span}\{1_{\mathbb{D}}, k\}.$$

We will call these *-subalgebras the **blue complex numbers**, **red complex numbers**, and **green complex numbers** respectively. A generic complex number will often be denoted by z . The adjugation operation corresponds to conjugation in the traditional sense over the complex numbers: if $z = x1_{\mathbb{D}} + yh \in \mathbb{C}_c(\mathbb{F})$ where h is one of i, j , or k , then

$$z^* = (x1_{\mathbb{D}} + yh)^* = x1_{\mathbb{D}} - yh. \tag{7}$$

Attempts to extend the notion of complex numbers are of course not new: a red complex number, also called a **split-complex number** or a **hyperbolic number**, was first referenced in the introduction of *tessarines* by Cockle [8], and appear in the split-biquaternions introduced by Clifford [9]. With our approach, formal symbols are replaced by specific matrices, which makes computations more explicit and provides a uniform view of all three blue, red, and green structures.

3.1 Geometry over complex numbers

Now we introduce some geometry into the algebra of dihedrons, and then restrict this to each of the colored complex numbers. This allows us to study a variety of geometrical situations that are parallel to familiar facts for complex numbers uniformly under the umbrella of the dihedron geometry.

Define a symmetric bilinear form $\langle \cdot, \cdot \rangle$ on $\mathbb{D} = \mathbb{D}(\mathbb{F}_q)$ by

$$\langle h_1, h_2 \rangle \equiv \frac{1}{2} \operatorname{tr}(h_1 h_2^*) \quad (8)$$

with associated quadratic form

$$Q(h) \equiv \langle h, h \rangle = \frac{1}{2} \operatorname{tr}(\det(h) 1_{\mathbb{D}}) = \det(h).$$

We call the number $Q(h)$ the **quadrance** of h , and note that $Q(h) = Q(h^*)$ for any dihedron h . A key consequence of the connection with the determinant is that for any dihedrons h_1 and h_2 we have

$$Q(h_1 h_2) = Q(h_1) Q(h_2). \quad (9)$$

Specifically the blue, red, and green geometries are obtained by restricting the symmetric bilinear form $\langle \cdot, \cdot \rangle$ and associated quadratic form Q on $\mathbb{D}$ to the respective $*$ -subalgebras $\mathbb{C}_c(\mathbb{F})$ for $c = b, r$, and g . If $z_1 = x_1 1_{\mathbb{D}} + y_1 h$ and $z_2 = x_2 1_{\mathbb{D}} + y_2 h$, with h either i, j or k in the three cases, then uniformly we may write

$$\langle z_1, z_2 \rangle = \frac{1}{2} \operatorname{tr}(z_1 z_2^*) = x_1 x_2 + \det(h) y_1 y_2.$$

So in the blue geometry, $\langle z_1, z_2 \rangle = x_1 x_2 + y_1 y_2$ with quadrance $Q(x 1_{\mathbb{D}} + y i) = x^2 + y^2$, while in the red and green geometries, $\langle z_1, z_2 \rangle = x_1 x_2 - y_1 y_2$ with quadrance $Q(x 1_{\mathbb{D}} + y j) = Q(x 1_{\mathbb{D}} + y k) = x^2 - y^2$. However in the green geometry we can also write the quadrance more simply as

$$Q \begin{pmatrix} t & 0 \\ 0 & u \end{pmatrix} = tu.$$

These formulas make it clear that the blue geometry can be identified with the familiar Euclidean geometry, while the red and green geometries correspond to two Einstein-Minkowski (relativistic) geometries. Physicists often refer to the green geometry in terms of null coordinates or light-cone coordinates in special relativity theory, by making a change of coordinate from the xy -coordinate to the tu -coordinate where $t = x + y$ and $u = x - y$ (see [29], [31], or [32]).

3.2 Unit circles in chromogeometry

For c one of b, r , or g , the **unit circle** $S_c(\mathbb{F}_q)$ is defined to be the set of complex numbers in $\mathbb{C}_{c,q}$ of quadrance 1_q , that is,

$$S_c(\mathbb{F}_q) \equiv \{z \in \mathbb{C}_{c,q} : Q(z) = 1_q\}.$$

From (9) we see that this set contains the identity $1_{\mathbb{D}}$, is closed under multiplication, and indeed forms a commutative group, since inverses are given by adjugates and in each of the three cases all elements are multiples of the identity and a single other matrix.

We identify the blue and red complex numbers $x1_{\mathbb{D}} + yi$ and $x1_{\mathbb{D}} + yj$ with the affine point $[x, y]$, and identify the green complex number $x1_{\mathbb{D}} + yk$ with the affine point $[x + y, x - y]$. With this identification, the unit circles $S_c(\mathbb{F}_q)$ can be realized as subsets of $\mathbb{A}$:

$$\begin{aligned} S_b(\mathbb{F}_q) &= \{[x, y] \in \mathbb{A} : x^2 + y^2 = 1\} \\ S_r(\mathbb{F}_q) &= \{[x, y] \in \mathbb{A} : x^2 - y^2 = 1\} \\ S_g(\mathbb{F}_q) &= \{[x, y] \in \mathbb{A} : xy = 1\}. \end{aligned}$$

There is also the shorthand notations $S_{b,q}$, $S_{r,q}$, and $S_{g,q}$ if the underlying field is clear from context, and $S_{c,q}$ to indicate the unit circle of any color. With this point of view, the blue unit circle is the unit circle in what is the finite field version of Euclidean geometry. The red and green unit circles can be regarded as two unit circles in finite field versions of the Minkowski-Einstein (relativistic) geometry.

While the green unit circle $S_{g,q}$ is easily parametrized algebraically, the traditional parametrizations for the blue and red unit circles over the real numbers are often presented via trigonometric and hyperbolic trigonometric functions. However, since our discussion is purely algebraic, we prefer to utilize the respective rational parametrizations

$$\left[\frac{1-t^2}{1+t^2}, \frac{2t}{1+t^2} \right] \quad \text{and} \quad \left[\frac{1+t^2}{1-t^2}, \frac{2t}{1-t^2} \right]$$

for the blue and red unit circles.

These parametrizations conveniently extend also to the finite field case. This is a classical argument, going back to at least Fermat in the Euclidean case, and arguably to Euclid from his classification of Pythagorean triples. Recall that for a prime power $q = p^r$, the **Jacobi symbol** may be defined by

$$\left(\frac{-1}{q} \right) = \left(\frac{-1}{p} \right)^r = \begin{cases} 1 & \text{if } -1 \text{ is a square in } \mathbb{F}_q \\ -1 & \text{if } -1 \text{ is not a square in } \mathbb{F}_q. \end{cases}$$

Lemma 3. *Every point on the green unit circle $S_{g,q}$ can be written as $[x, y] = [t, t^{-1}]$ for a unique element $t \in \mathbb{F}_q^\times$. Every point on the red unit circle $S_{r,q}$ except for $[-1, 0]$, can be written as $[x, y] = \left[\frac{1+t^2}{1-t^2}, \frac{2t}{1-t^2} \right]$ for a unique $t \neq \pm 1$. Every point on the blue unit circle $S_{b,q}$, except for $[-1, 0]$, can be written as $[x, y] = \left[\frac{1-t^2}{1+t^2}, \frac{2t}{1+t^2} \right]$ for a unique t satisfying $t^2 \neq -1$. In particular*

$$|S_{g,q}| = q - 1 \quad |S_{r,q}| = q - 1 \quad |S_{b,q}| = q - \left(\frac{-1}{q} \right).$$

Proof. The parametrization of $S_{g,q}$ is evident with $|S_{g,q}| = |\mathbb{F}_q^\times| = q - 1$. On $S_{r,q}$, the identity

$$\left(\frac{1+t^2}{1-t^2} \right)^2 - \left(\frac{2t}{1-t^2} \right)^2 = 1$$

holds for all $t \in \mathbb{F}_q$ for which $1 - t^2 \neq 0$. The line through the points $[-1, 0]$ and $[0, t]$ given by $y = tx + t$ intersects the red unit circle $x^2 - y^2 = 1$ in exactly two points, one of which is $[-1, 0]$, and the other, by solving $x^2 - (tx + t)^2 = 1$ for x is $\left[\frac{1+t^2}{1-t^2}, \frac{2t}{1-t^2} \right]$. Hence every point on $S_{r,q}$ except $[-1, 0]$ corresponds to exactly one $t \in \mathbb{F}_q - \{\pm 1\}$, and so $|S_{r,q}| = 1 + (q - 2) = q - 1$.

The parametrization of the blue unit circle is done similarly, with a factor $1+t^2$ in the denominator, so we have to exclude those $t \in \mathbb{F}_q$ whose square is -1 , if any. If -1 is not a square then the number of such points is $q+1$. Otherwise, there are exactly two solutions to the equation $t^2 = -1$, so the number of points on $S_{b,q}$ is $1+(q-2) = q-1$. $\square$

There is also additional algebraic structure to $\mathbb{C}_{b,q}$ that holds only when -1 is not a square. In this case $\mathbb{C}_{b,q}$ is a field of size q^2 , since an element $z = x1_{\mathbb{D}} + yi \in \mathbb{C}_{b,q}$ is non-invertible precisely when $Q(z) = x^2 + y^2 = \det(z) = 0$, and the only solution is $x = y = 0$, since if either one of them is non-zero, say y , then $y^2 \neq 0$ and therefore $\left(\frac{x}{y}\right)^2 + 1 = 0$ which is impossible.

Proposition 3. *If -1 is a square in $\mathbb{F}_q$ then $S_{g,q}$, $S_{r,q}$, and $S_{b,q}$ are each cyclic groups of order $q-1$, hence isomorphic. When -1 is not a square in $\mathbb{F}_q$, $S_{b,q}$ is a cyclic group of order $q+1$.*

Proof. By Lemma 3, any point on $S_{g,q}$ can be written as $[t, t^{-1}]$ for $t \in \mathbb{F}_q^\times$. The map $t \mapsto [t, t^{-1}]$ from $\mathbb{F}_q^\times$ to $S_{g,q}$ is an isomorphism and since $\mathbb{F}_q^\times$ is cyclic of order $q-1$, so is $S_{g,q}$. Now $S_{g,q}$ and $S_{r,q}$ are isomorphic through the map

$$\theta : S_r(\mathbb{F}_q) \longrightarrow S_g(\mathbb{F}_q) \quad \text{where} \quad x1_{\mathbb{D}} + yj \mapsto x1_{\mathbb{D}} + yk.$$

When $-1 = i^2$ for some $i \in \mathbb{F}_q$, the isomorphism between $S_{r,q}$ and $S_{b,q}$ is established through the map

$$\varrho : S_b(\mathbb{F}_q) \longrightarrow S_r(\mathbb{F}_q) \quad \text{where} \quad x1_{\mathbb{D}} + yi \mapsto x1_{\mathbb{D}} + yj.$$

Therefore the three unit circles are cyclic groups of order $q-1$.

Now if -1 is not a square in $\mathbb{F}_q$, we may extend $\mathbb{F}_q$ to the quadratic field extension $\mathbb{C}_{b,q}$ and regard $S_{b,q}$ as a subgroup of $\mathbb{C}_{b,q}^\times$, hence also cyclic. The order of $S_{b,q}$ is established by Lemma 3. $\square$

This cyclic group structure of all of the unit circles has a particularly pleasant property that will become important later on. We shall state the result more generally.

Proposition 4. *Let $\mathbb{F}_q$ be a finite field of size q and $\mathbb{F}_q^\times$ be its multiplicative group. Suppose C is a cyclic subgroup of $\mathbb{F}_q^\times$ of order n . Then*

$$\sum_{t \in C} t^k = \begin{cases} n1_q & \text{if } n \mid k \\ 0 & \text{otherwise.} \end{cases}$$

Proof. Let C be generated by $\zeta \neq 1_q$, so $C = \{1_q, \zeta, \zeta^2, \dots, \zeta^{n-1}\}$ with $\zeta^n = 1_q$. If $k = mn$ for some integer m , then

$$\begin{aligned} \sum_{t \in C} t^k &= 1_q^{mn} + \zeta^{mn} + \zeta^{2mn} + \dots + \zeta^{(n-1)mn} \\ &= \underbrace{1_q + 1_q + \dots + 1_q}_{n \text{ times}} = n1_q. \end{aligned}$$

If n does not divide k , then $\zeta^k \neq 1_q$ and so $\sum_{t \in C} t^k = \sum_{l=0}^{n-1} \zeta^{kl} = \frac{\zeta^{kn} - 1_q}{\zeta^k - 1_q} = 0$ since $\zeta^{kn} = (\zeta^n)^k = 1_q$ and $\zeta^k \neq 1_q$. $\square$

3.3 Special orthogonal groups in chromogeometry

The three different geometries on $\mathbb{C}_{c,q}$ induced by the symmetric bilinear form (8) correspond to the green, red, and blue geometries on $\mathbb{A}$ by the identification stipulated previously: $x1_{\mathbb{D}} + yi$ and $x1_{\mathbb{D}} + yj$ as $[x, y]$, and $x1_{\mathbb{D}} + yk$ as $[x + y, x - y]$.

More specifically, over the green geometry, for $z_1 = x_1 1_{\mathbb{D}} + y_1 k$ and $z_2 = x_2 1_{\mathbb{D}} + y_2 k \in \mathbb{C}_{g,q}$ we have

$$\langle z_1, z_2 \rangle = \frac{1}{2} \text{tr}(z_1 z_2^*) = \frac{1}{2} ((x_1 + y_1)(x_2 - y_2) + (x_1 - y_1)(x_2 + y_2))$$

so the corresponding symmetric bilinear form on $\mathbb{A}$ is

$$[x_1, y_1] \cdot_g [x_2, y_2] \equiv \frac{1}{2} (x_1 y_2 + x_2 y_1).$$

We call the geometry induced by $\cdot_g$ the green geometry on $\mathbb{A}$.

Similarly, we have the red and blue geometry on $\mathbb{A}$ defined by the symmetric bilinear forms

$$\begin{aligned} [x_1, y_1] \cdot_r [x_2, y_2] &\equiv x_1 x_2 - y_1 y_2 \\ [x_1, y_1] \cdot_b [x_2, y_2] &\equiv x_1 x_2 + y_1 y_2. \end{aligned}$$

A simple algebraic manipulation shows that $[x_1, y_1] \cdot_c [x_2, y_2] = (x_1, y_1) M_c (x_2, y_2)^T$ where M_c are the invertible matrices

$$M_g = \begin{pmatrix} 0 & \frac{1}{2} \\ \frac{1}{2} & 0 \end{pmatrix} \quad M_r = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \quad M_b = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}. \quad (10)$$

If $\text{GL}(2, \mathbb{F}_q)$ denotes the group of 2×2 invertible matrices over $\mathbb{F}_q$, there is a natural right action of $\text{GL}(2, \mathbb{F}_q)$ on $\mathbb{A}$ and left action of $\text{GL}(2, \mathbb{F}_q)$ on $\text{Pol}_2(\mathbb{F}_q)$ as

$$\begin{aligned} [x, y] \cdot \begin{pmatrix} h_{11} & h_{12} \\ h_{21} & h_{22} \end{pmatrix} &\equiv [h_{11}x + h_{21}y, h_{12}x + h_{22}y] \\ \begin{pmatrix} h_{11} & h_{12} \\ h_{21} & h_{22} \end{pmatrix} \cdot \pi &\equiv \pi(h_{11}\alpha + h_{21}\beta, h_{12}\alpha + h_{22}\beta) \end{aligned}$$

where we regard π as a polynomial in α and β . The subgroup of $\text{GL}(2, \mathbb{F}_q)$ consisting of matrices h that preserve that bilinear form $\cdot_c$ is called the **orthogonal group** $O_c(\mathbb{F}_q) = O_{c,q}$ with respect to geometry c . Any matrix $h \in O_{c,q}$ necessarily satisfies $h M_c h^T = M_c$ where M_c is given in (10) and therefore $\det(h) = \pm 1$.

The collection of matrices with determinant 1 in $O_{c,q}$ is itself a group which we call the **rotation group** $SO_c(\mathbb{F}_q) = SO_{c,q}$. We are particularly interested in the analysis pertaining to the rotation group $SO_{c,q}$ by considering how $SO_{c,q}$ acts on various spaces. $SO_{c,q}$ naturally right acts on $\mathbb{A}$ via the restriction of the right action of $\text{GL}(2, \mathbb{F}_q)$ on $\mathbb{A}$. There is then an associated left action of $SO_{c,q}$ on $\mathbb{F}_q^{\mathbb{A}}$ and an associated right action of $SO_{c,q}$ on $(\mathbb{F}_q^{\mathbb{A}})^*$. Similarly, the left action of $SO_{c,q}$ on $\text{Pol}_2(\mathbb{F}_q)$ is induced as the restriction of the action of $\text{GL}(2, \mathbb{F}_q)$ on $\text{Pol}_2(\mathbb{F}_q)$. This action respects evaluation in a sense that

$$\varepsilon(h \cdot \pi) = h \cdot \varepsilon(\pi). \quad (11)$$

This setup now makes it easy and pleasant to explicitly describe the rotation group $SO_{c,q}$.

Theorem 8. *The rotation group $SO_{c,q}$ is precisely the unit circle group $S_{c,q}$.*

Proof. Any $h \in SO_{c,q}$ satisfies $\det(h) = 1$ by definition. Since over any geometry c , h must be in $\mathbb{C}_{c,q}$, the condition $\det(h) = 1$ is equivalent to $h \in S_{c,q}$. This implies that $SO_{c,q} \subseteq S_{c,q}$.

Similarly, if $z \in S_{c,q}$ then $Q(z) = \det(z) = 1$ by definition. The condition that z preserves the bilinear form $\cdot_c$ on $\mathbb{A}$ is equivalent to $zM_c z^T = M_c$. Over the green geometry, z has the form $x1_{\mathbb{D}} + yk$ where $x^2 - y^2 = 1$ and so by inspection,

$$zM_g z^T = \begin{pmatrix} 0 & \frac{1}{2}(x^2 - y^2) \\ \frac{1}{2}(x^2 - y^2) & 0 \end{pmatrix} = \begin{pmatrix} 0 & \frac{1}{2} \\ \frac{1}{2} & 0 \end{pmatrix} = M_g.$$

The cases $z \in S_{r,q}$ and $z \in S_{b,q}$ are treated similarly. In all cases, $zM_c z^T = M_c$ and thus z preserves the bilinear form $\cdot_c$ on $\mathbb{A}$. This shows that $S_{c,q} \subseteq SO_{c,q}$. $\square$

The result above allows us to present the following observation about the action of $SO_{c,q}$ on $\mathbb{A}$.

Theorem 9. *The unit circle $S_{c,q}$ is an orbit of the group $SO_{c,q}$ in $\mathbb{A}$. Moreover, the action of $SO_{c,q}$ in $S_{c,q}$ is faithful.*

Proof. Since $S_{c,q} = SO_{c,q}$ by Theorem 8, we may identify $[x, y] \cdot h$ as zh where $z \in \mathbb{C}_{c,q}$ is the complex number associated to $[x, y]$. This allows us to observe that $[x, y] \cdot h \in S_{c,q}$ for any $h \in SO_{c,q}$. To see that the action of $SO_{c,q}$ in $S_{c,q}$ is faithful, the equation $[x, y] \cdot h = [x, y]$ for any $[x, y] \in S_{c,q}$ may be identified as $zh = z$ for all $z \in S_{c,q}$. This yields $h = 1_{\mathbb{D}}$, finishing our claim. $\square$

As a final remark, the theory of orthogonal matrices over finite fields over a general geometry has been intensively studied (see [5], [24], [35]). The size of the group $SO_b(n, \mathbb{F}_q)$ consisting of all $n \times n$ orthogonal matrices over $\mathbb{F}_q$ with determinant 1 is given by

$$|SO_b(2m+1, \mathbb{F}_q)| = q^{m^2} \prod_{k=1}^m (q^{2k} - 1)$$

and

$$|SO_b(2m, \mathbb{F}_q)| = \begin{cases} \frac{1}{q^m + 1} \prod_{k=0}^{m-1} (q^{2m} - q^{2k}) & \text{if } \left(\frac{-1}{q}\right) = 1 \\ \frac{1}{q^m + (-1)^m} \prod_{k=0}^{m-1} (q^{2m} - q^{2k}) & \text{if } \left(\frac{-1}{q}\right) = -1. \end{cases}$$

It is clear that when $n = 2$, the formula above matches with the result established in Theorem 8.

4 Circular Integral Functionals

We now introduce an algebraic integral on the unit circles in each of the three geometries, valid over a general finite field $\mathbb{F}_q$ of odd characteristic. These will be functionals on polynomials which satisfies conditions that we call *normalization*, *locality* and *invariance*. We begin with the simpler case of a corresponding functional on the space $\mathbb{F}_q^{\mathbb{A}}$ of $\mathbb{F}_q$ -valued functions on $\mathbb{A}$.

4.1 Circular integral functionals on $\mathbb{F}_q^\mathbb{A}$

For one of the three geometries $c = b, r, \text{ or } g$ on the affine space $\mathbb{A} = \mathbb{A}(\mathbb{F}_q)$, the unit circle is $S_{c,q}$ and the special orthogonal group is $SO_{c,q}$. A linear functional ϕ on $\mathbb{F}_q^\mathbb{A}$ is a **circular integral functional** precisely when the following three conditions are satisfied:

(Normalization) The functional ϕ maps the constant function $1_{\mathbb{F}^\mathbb{A}}$ to 1_q

(Locality) If $f \in \mathbb{F}_q^\mathbb{A}$ restricts to the zero function on $S_{c,q}$, then $\phi(f) = 0$

(Invariance) The functional ϕ is $SO_{c,q}$ -invariant, that is $\phi \cdot h = \phi$ for all $h \in SO_{c,q}$.

Theorem 10. *There is one and only one circular integral functional on $\mathbb{F}_q^\mathbb{A}$, namely*

$$\phi(f) = \frac{1}{|S_{c,q}|} \sum_{[x,y] \in S_{c,q}} f(x, y).$$

Proof. The definition of ϕ above is well-defined since $|S_{c,q}| \not\equiv 0 \pmod{p}$ by Proposition 3. This functional is normalized, since

$$\phi(1_{\mathbb{F}^\mathbb{A}}) = \frac{1}{|S_{c,q}|} \sum_{[x,y] \in S_{c,q}} 1_q = 1_q$$

and from the definition it is clearly localized. For the Invariance property, for $h \in SO_{c,q}$ and $f \in \mathbb{F}_q^\mathbb{A}$, we have $(\phi \cdot h)(f) = \phi(h \cdot f)$ and therefore

$$(\phi \cdot h)(f) = \frac{1}{|S_{c,q}|} \sum_{[x,y] \in S_{c,q}} (h \cdot f)(x, y) = \frac{1}{|S_{c,q}|} \sum_{[x,y] \in S_{c,q}} f([x, y] \cdot h) = \frac{1}{|S_{c,q}|} \sum_{[x,y] \in S_{c,q}} f(x, y) = \phi(f)$$

since $SO_{c,q}$ acts as a group of bijections on $S_{c,q}$. Consequently ϕ is a circular integral functional.

Now if η is any circular integral functional on $\mathbb{F}_q^\mathbb{A}$, since Δ is a basis for $\mathbb{F}_q^\mathbb{A}$, it suffices to show that $\eta(\delta_{[a,b]}) = \phi(\delta_{[a,b]})$ for all $[a, b] \in \mathbb{A}$. We observe that

$$\phi(\delta_{[a,b]}) = \frac{1}{|S_{c,q}|} \sum_{[x,y] \in S_{c,q}} \delta_{[a,b]}(x, y) = \begin{cases} \frac{1}{|S_{c,q}|} & \text{if } [a, b] \in S_{c,q} \\ 0 & \text{if } [a, b] \notin S_{c,q}. \end{cases} \quad (12)$$

If $[a, b] \notin S_{c,q}$, then $\delta_{[a,b]} = 0$ on $S_{c,q}$ and therefore by Locality, $\eta(\delta_{[a,b]}) = 0$. On the other hand if $[a, b] \in S_{c,q}$, then by Invariance, for any $h \in SO_{c,q}$ it follows that

$$\eta(\delta_{[a,b]}) = \eta(h \cdot \delta_{[a,b]}) = \eta(\delta_{[a,b] \cdot h}). \quad (13)$$

Since $SO_{c,q}$ acts transitively on $S_{c,q}$, (13) implies that η is constant on the set $\Delta|_{S_{c,q}}$. Consequently,

$$1_q = \eta(1_{\mathbb{F}^\mathbb{A}}) = \eta\left(\sum_{[x,y] \in \mathbb{A}} \delta_{[x,y]}\right) = \eta\left(\sum_{[x,y] \in S_{c,q}} \delta_{[x,y]}\right) = \sum_{[x,y] \in S_{c,q}} \eta(\delta_{[x,y]}) = |S_{c,q}| \eta(\delta_{[a,b]})$$

where the second equality comes from (1). This concludes the proof. $\square$

4.2 Circular integral functionals on $\text{Pol}_2(\mathbb{F}_q)$

A linear functional ψ on $\text{Pol}_2(\mathbb{F}_q)$ is called a **circular integral functional** precisely when the following three conditions are satisfied:

(Normalization) The functional ψ maps the polynumber $\mathbf{1}$ to 1_q .

(Locality) If $\pi \in \text{Pol}_2(\mathbb{F}_q)$ evaluates to the zero function on $S_{c,q}$ then $\psi(\pi) = 0$.

(Invariance) The functional ψ is $SO_{c,q}$ -invariant, that is $\psi(h \cdot \pi) = \psi(\pi)$ for any $\pi \in \text{Pol}_2(\mathbb{F}_q)$ and $h \in SO_{c,q}$.

These three conditions are virtually identical to the corresponding conditions for a circular integral functional on $\mathbb{F}_q^{\mathbb{A}}$, however there is quite a subtle difference in the analysis. Recall that $\varepsilon(\pi)$ refers to the evaluation of a polynumber π .

Theorem 11. *There is one and only one circular integral functional on $\text{Pol}_2(\mathbb{F}_q)$ namely*

$$\psi_{c,q}(\pi) = \frac{1}{|S_{c,q}|} \sum_{[x,y] \in S_{c,q}} \varepsilon(\pi)(x, y).$$

Proof. We first note that $\psi_{c,q} = \phi \circ \varepsilon$ where ϕ is the unique circular integral functional on $\mathbb{F}_q^{\mathbb{A}}$ established in Theorem 10. Since $\varepsilon(\mathbf{1}) = 1_{\mathbb{F}^{\mathbb{A}}}$, the Normalization condition follows from that of ϕ . The Locality property is also obvious, for if $\pi \in \text{Pol}_2(\mathbb{F}_q)$ evaluates to the zero function on $S_{c,q}$, then $\varepsilon(\pi)(x, y) = 0$ for all $[x, y] \in S_{c,q}$.

To prove the Invariance property, we note from (11) that the $SO_{c,q}$ -left-action on $\text{Pol}_2(\mathbb{F}_q)$ respects evaluation, so

$$\begin{aligned} \psi_{c,q}(h \cdot \pi) &= \frac{1}{|S_{c,q}|} \sum_{[x,y] \in S_{c,q}} \varepsilon(h \cdot \pi)(x, y) = \frac{1}{|S_{c,q}|} \sum_{[x,y] \in S_{c,q}} (h \cdot \varepsilon(\pi))(x, y) \\ &= \phi(h \cdot \varepsilon(\pi)) = \phi(\varepsilon(\pi)) = \psi_{c,q}(\pi) \end{aligned}$$

where we used the Invariance property of ϕ . We deduce that $\psi_{c,q}$ is a circular integral functional on $\text{Pol}_2(\mathbb{F}_q)$ with respect to $S_{c,q}$.

Now if χ is any circular integral functional on $\text{Pol}_2(\mathbb{F}_q)$ with respect to $S_{c,q}$, we need to show that $\chi(\pi) = \psi_{c,q}(\pi)$ for any $\pi \in \text{Pol}_2(\mathbb{F}_q)$. By Theorem 1, the decomposition $\pi = \pi_1 + \pi_2$ where $\pi_1 \in \mathcal{P}_q$ and $\pi_2 \in \ker(\varepsilon)$ is unique. Since $\varepsilon(\pi_2) = 0$, Locality asserts that $\chi(\pi_2) = 0$. This means that we only need to consider the possible values of χ on $\mathcal{P}_q$.

Since Π is a basis for $\mathcal{P}_q$, it suffices to show that

$$\chi(\pi_{[a,b]}) = \psi_{c,q}(\pi_{[a,b]}) = \begin{cases} \frac{1}{|S_{c,q}|} & \text{if } [a, b] \in S_{c,q} \\ 0 & \text{if } [a, b] \notin S_{c,q} \end{cases}$$

where the second equality follows from (12).

If $[a, b] \notin S_{c,q}$, then $\varepsilon(\pi_{[a,b]}) = \delta_{[a,b]} = 0$ on $S_{c,q}$ which implies that $\chi(\pi_{[a,b]}) = 0$ by Locality. Now suppose $[a, b] \in S_{c,q}$. In this case,

$$\varepsilon(h \cdot \pi_{[a,b]}) = h \cdot \varepsilon(\pi_{[a,b]}) = h \cdot \delta_{[a,b]} = \delta_{[a,b] \cdot h^{-1}} = \varepsilon(\pi_{[a,b] \cdot h^{-1}})$$

where the third equality above follows from equation (2). It follows that $h \cdot \pi_{[a,b]} - \pi_{[a,b] \cdot h^{-1}} \in \ker(\varepsilon)$. By Invariance and Locality,

$$\chi(\pi_{[a,b] \cdot h^{-1}}) = \chi(h \cdot \pi_{[a,b]}) = \chi(\pi_{[a,b]}).$$

This implies that χ is constant on $\Pi|_{S_{c,q}}$.

Finally,

$$1_q = \chi(\mathbf{1}) = \chi\left(\sum_{[a,b] \in \mathbb{A}} \pi_{[a,b]}\right) = \chi\left(\sum_{[a,b] \in S_{c,q}} \pi_{[a,b]}\right) = \sum_{[a,b] \in S_{c,q}} \chi(\pi_{[a,b]}) = |S_{c,q}| \chi(\pi_{[a,b]})$$

which concludes the proof. $\square$

We call the circular integral functionals $\psi_{c,q}$ the **green**, **red**, or **blue Fourier summation functionals** respectively. Since $|S_{c,q}|$ is known, we obtain the following three formulas:

$$\psi_{g,q}(\alpha^k \beta^l) \equiv - \sum_{[x,y] \in S_{g,q}} x^k y^l \quad \psi_{r,q}(\alpha^k \beta^l) \equiv - \sum_{[x,y] \in S_{r,q}} x^k y^l \quad \psi_{b,q}(\alpha^k \beta^l) \equiv - \left(\frac{-1}{q}\right) \sum_{[x,y] \in S_{b,q}} x^k y^l.$$

We give two examples of the computation of the Fourier summation functionals $\psi_{c,q}$.

Example 3. Over $\mathbb{F}_{17} = \{0, 1, 2, \dots, 15, 16\}$, the red unit circle is

$$S_{r,17} = \left\{ \begin{array}{cccccccc} [1, 0] & [16, 0] & [6, 1] & [6, 16] & [11, 1] & [11, 16] & [3, 5] & [3, 12] \\ [14, 5] & [14, 12] & [4, 7] & [4, 10] & [13, 7] & [13, 10] & [0, 4] & [0, 13] \end{array} \right\}.$$

It can be verified that, for example,

$$\begin{aligned} \psi_{r,17}(\alpha^6 \beta^4) &= - \sum_{[x,y] \in S_{r,17}} x^6 y^4 \\ &= - \left(\begin{array}{l} 1^6 0^4 + 16^6 0^4 + 0^6 1^4 + 0^6 16^4 + 4^6 6^4 + 4^6 11^4 + 13^6 6^4 + 13^6 11^4 + \\ 6^6 4^4 + 6^6 13^4 + 11^6 4^4 + 11^6 13^4 + 3^6 3^4 + 3^6 14^4 + 14^6 3^4 + 14^6 14^4 \end{array} \right) = 3. \end{aligned}$$

Example 4. Since $1 + \alpha - \alpha^3$ is irreducible in $\text{Pol}(\mathbb{F}_3)$, we may regard $\mathbb{F}_{27}$ as $\text{Pol}(\mathbb{F}_3) / (1 + \alpha - \alpha^3)$. By identifying $\pi + (1 + \alpha - \alpha^3)$ simply as π , the blue unit circle is

$$S_{b,27} = \left\{ \begin{array}{cccccccc} [1, 0] & [2, 0] & [0, 1] & [0, 2] & [\alpha^2, 2\alpha + \alpha^2] & [\alpha^2, \alpha + 2\alpha^2] & [2\alpha^2, 2\alpha + \alpha^2] \\ [2\alpha^2, \alpha + 2\alpha^2] & [2\alpha + \alpha^2, \alpha^2] & [\alpha + 2\alpha^2, \alpha^2] & [2\alpha + \alpha^2, 2\alpha^2] \\ [\alpha + 2\alpha^2, 2\alpha^2] & [2 + \alpha^2, 1 + \alpha + \alpha^2] & [2 + \alpha^2, 2 + 2\alpha + 2\alpha^2] \\ [2\alpha + 2\alpha^2, 1 + 2\alpha + \alpha^2] & [1 + 2\alpha^2, 2 + 2\alpha + 2\alpha^2] \\ [1 + 2\alpha^2, 1 + \alpha + \alpha^2] & [1 + \alpha + \alpha^2, 2 + \alpha^2] & [2 + 2\alpha + 2\alpha^2, 2 + \alpha^2] \\ [1 + \alpha + \alpha^2, 1 + 2\alpha^2] & [\alpha + \alpha^2, 1 + 2\alpha + \alpha^2] & [\alpha + \alpha^2, 2 + \alpha + 2\alpha^2] \\ [2\alpha + 2\alpha^2, 2 + \alpha + 2\alpha^2] & [1 + 2\alpha + \alpha^2, \alpha + \alpha^2] & [2 + \alpha + 2\alpha^2, \alpha + \alpha^2] \\ [2 + 2\alpha + 2\alpha^2, 1 + 2\alpha^2] & [1 + 2\alpha + \alpha^2, 2\alpha + 2\alpha^2] & [2 + \alpha + 2\alpha^2, 2\alpha + 2\alpha^2] \end{array} \right\}.$$

Using this we may directly compute that

$$\psi_{b,27}(\alpha^4\beta^6) = -\left(\frac{-1}{27}\right) \sum_{[x,y] \in S_{b,27}} x^4 y^6 = -\left(\frac{-1}{3}\right)^3 \sum_{[x,y] \in S_{b,27}} x^4 y^6 = 0.$$

5 Explicit Formulas for $\psi_{c,q}$

The computation of $\psi_{c,q}$ can become imposing, especially when the size of the field is large. The second half of this paper develops explicit formulas that turn out to involve the circular super Catalan numbers in the red and blue geometries. A formula for the green Fourier summation functional $\psi_{g,q}$ is relatively easy to find, and will be the key to obtaining formulas for $\psi_{r,q}$, and for $\psi_{b,q}$ in the case when -1 is a square in $\mathbb{F}_q$. This chain of connection suggests that relativistic geometries might be simpler than Euclidean geometry, at least in the analysis context of integration theories over general fields.

The formula for $\psi_{b,q}$ when -1 is not a square in $\mathbb{F}_q$ warrants a special treatment involving Fourier analysis style arguments that are closer to the case for the field of real numbers, in which -1 is also not a square.

Theorem 12. *In the green geometry over $\mathbb{F}_q$, for any $k, l \in \mathbb{N}$,*

$$\psi_{g,q}(\alpha^k \beta^l) = \begin{cases} 1_q & \text{if } |S_{g,q}| \mid (k-l), \\ 0 & \text{otherwise.} \end{cases}$$

Proof. From the parametrization of $S_{g,q}$, we observe that

$$\psi_{g,q}(\alpha^k \beta^l) = - \sum_{[x,y] \in S_{g,q}} x^k y^l = - \sum_{t \in \mathbb{F}_q^\times} t^k (t^{-1})^l = - \sum_{t \in \mathbb{F}_q^\times} t^{k-l}.$$

The conclusion follows by applying Lemma 4 with $C = \mathbb{F}_q^\times$ since $q-1$ equals $|S_{g,q}|$. $\square$

An explicit formula for $\psi_{r,q}$ can then be found by utilizing the isomorphism θ between $S_{r,q}$ and $S_{g,q}$, and involves two numerical invariants along with the coefficients of the circular polynoms $\pi_{k,l}$.

Theorem 13. *In the red geometry over $\mathbb{F}_q$, define $w \equiv \frac{1}{2} |S_{b,q}| = \frac{1}{2} (q-1)$, and for natural numbers k and l , let $R_{k,l} \equiv \left\lfloor \frac{k+l}{|S_{b,q}|} \right\rfloor$. Then*

$$\psi_{r,q}(\alpha^k \beta^l) = \begin{cases} \left(\sum_{|d| \leq R_{k,l}} [\alpha^{m+n+dw}] \right) \pi_{k,l} \bmod p & \text{if } k = 2m, l = 2n \\ 0 & \text{otherwise.} \end{cases}$$

Proof. For any $[x, y] \in S_{r,q}$, write $u = x + y$ and $v = x - y$ so that $[u, v] \in S_{g,q}$. Inversely the expressions $x = \frac{u+v}{2}$ and $y = \frac{u-v}{2}$ are both well-defined since $2 \neq 0$. Using the definition of $\psi_{r,q}$,

$$\psi_{r,q}(\alpha^k \beta^l) = - \sum_{[x,y] \in S_{r,q}} x^k y^l = - \frac{1}{2^{k+l}} \sum_{[u,v] \in S_{g,q}} (u+v)^k (u-v)^l.$$

Reverting back to x and y , we obtain a formula for $\psi_{r,q}$ in terms of $\psi_{g,q}$ as follows:

$$\begin{aligned}
\psi_{r,q}(\alpha^k \beta^l) &= -\frac{1}{2^{k+l}} \sum_{[x,y] \in S_{g,q}} \left(\sum_{s=0}^k \binom{k}{s} x^s y^{k-s} \right) \left(\sum_{t=0}^l (-1)^{l-t} \binom{l}{t} x^t y^{l-t} \right) \\
&= -\frac{1}{2^{k+l}} \sum_{s=0}^k \sum_{t=0}^l (-1)^{l-t} \binom{k}{s} \binom{l}{t} \sum_{[x,y] \in S_{g,q}} x^{s+t} y^{k+l-(s+t)} \\
&= \frac{(-1)^l}{2^{k+l}} \sum_{s=0}^k \sum_{t=0}^l (-1)^t \binom{k}{s} \binom{l}{t} \psi_{g,q}(\alpha^{s+t} \beta^{k+l-(s+t)})
\end{aligned} \tag{14}$$

where the minus sign originally in front of the expression has been absorbed.

By Theorem 12, $\psi_{g,q}(\alpha^{s+t} \beta^{k+l-(s+t)}) = 1_q$ precisely when

$$2(s+t) = k+l+d(q-1) \tag{15}$$

for some $d \in \mathbb{Z}$ and 0 otherwise. Note that here s and t are indices in the ranges $0 \leq s \leq k$ and $0 \leq t \leq l$.

If $k+l$ is odd, then by comparing the parity of both sides, (15) has no solution for s and t . We conclude that $\psi_{g,q}(\alpha^{s+t} \beta^{k+l-(s+t)}) = 0$ in this case, which leads to $\psi_{r,q}(\alpha^k \beta^l) = 0$. If $k+l = 2N$ for some integer N , (15) has solutions for d precisely when $|d| \leq R_{k,l}$ so that (15) reduces to $s+t = N+dw$ where $w = \frac{1}{2}(q-1)$.

Consequently, we can simplify the right-hand side of (14) to

$$\psi_{r,q}(\alpha^k \beta^l) = \frac{(-1)^l}{2^{2N}} \sum_{|d| \leq R_{k,l}} \sum_{s=0}^k (-1)^{N+dw-s} \binom{k}{s} \binom{l}{N+dw-s} 1_q. \tag{16}$$

For each d ,

$$\sum_{s=0}^k (-1)^{N+dw-s} \binom{k}{s} \binom{l}{N+dw-s} = [\alpha^{N+dw}] (1+\alpha)^k (1-\alpha)^l$$

so (16) becomes

$$\psi_{r,q}(\alpha^k \beta^l) = \frac{(-1)^l}{2^{2N}} \sum_{|d| \leq R_{k,l}} [\alpha^{N+dw}] (1+\alpha)^k (1-\alpha)^l 1_q. \tag{17}$$

Using Lemma 2,

$$[\alpha^{N+dw}] (1+\alpha)^k (1-\alpha)^l = (-1)^k [\alpha^{N-dw}] (1+\alpha)^k (1-\alpha)^l$$

for all $|d| \leq R_{k,l}$ so if k is odd, (17) simplifies to $\psi_{r,q}(\alpha^k \beta^l) = 0$.

We may therefore write $k = 2m$ and $l = 2n$. In this case, $N = m+n$ and (17) becomes

$$\begin{aligned}
\psi_{r,q}(\alpha^k \beta^l) &= \frac{1}{2^{2m+2n}} \sum_{|d| \leq R_{k,l}} [\alpha^{m+n+dw}] (1+\alpha)^{2m} (1-\alpha)^{2n} 1_q \\
&= \left(\sum_{|d| \leq R_{k,l}} [\alpha^{m+n+dw}] \right) \pi_{2m,2n} \bmod p.
\end{aligned}$$

□

Here is the special case that brings in the circular super Catalan numbers.

Theorem 14. *In the red geometry over $\mathbb{F}_q$ where $q = p^r$ for some prime p , suppose $k, l \in \mathbb{N}$ such that $0 \leq k + l < q - 1$. Then*

$$\psi_{r,q}(\alpha^k \beta^l) = (-1)^n \Omega(m, n) \bmod p.$$

Proof. From Theorem 13 we deduce that if $0 \leq k + l < q - 1$ then $R_{k,l} = 0$ and so

$$\begin{aligned} \psi_{r,q}(\alpha^k \beta^l) &= [\alpha^{m+n}] \pi_{2m,2n} \bmod p = (-1)^n [\alpha^{m+n}] (-1)^n \pi_{2m,2n} \bmod p \\ &= (-1)^n \Omega(m, n) \bmod p \end{aligned}$$

where the last equation follows from the restatement of the result of Georgiadis, Munemasa, and Tanaka [14] given as Theorem 7. $\square$

Now suppose $-1 = \iota^2$ for some element $\iota \in \mathbb{F}_q$. From Proposition 3, the red and blue unit circles are isomorphic groups, and so we may obtain an explicit formula for $\psi_{b,q}$ which is a small tweak from that for $\psi_{r,q}$.

Theorem 15. *In the blue geometry over $\mathbb{F}_q$ when -1 is a square, define $w \equiv \frac{1}{2} |S_{b,q}| = \frac{1}{2} (q - 1)$ and for natural numbers k and l , let $R_{k,l} \equiv \left\lfloor \frac{k+l}{|S_{b,q}|} \right\rfloor$. Then*

$$\psi_{b,q}(\alpha^k \beta^l) = \begin{cases} \left(\sum_{|d| \leq R_{k,l}} [\alpha^{m+n+dw}] \right) (-1)^n \pi_{k,l} \bmod p & \text{if } k = 2m, l = 2n \\ 0 & \text{otherwise.} \end{cases}$$

Proof. Using the bijection ϱ in the proof of Proposition 3, we determine that

$$\psi_{b,q}(\alpha^k \beta^l) = - \sum_{[x,y] \in S_{b,q}} x^k y^l = -(-\iota)^l \sum_{[x,y\iota] \in S_{r,q}} x^k (y\iota)^l.$$

By renaming the point $[x, y\iota] \in S_r(\mathbb{F}_q)$ to $[x, y]$,

$$\psi_{b,q}(\alpha^k \beta^l) = -(-\iota)^l \sum_{[x,y\iota] \in S_{r,q}} x^k (y\iota)^l = -(-\iota)^l \sum_{[x,y] \in S_{r,q}} x^k y^l = (-\iota)^l \psi_{r,q}(\alpha^k \beta^l).$$

The result then follows from the formula of $\psi_{r,q}(\alpha^k \beta^l)$ given in Theorem 13 above. If either k or l is odd, then $\psi_{b,q}(\alpha^k \beta^l) = 0$, and if $k = 2m$ and $l = 2n$, then

$$\psi_{b,q}(\alpha^k \beta^l) = (-\iota)^{2n} \psi_{r,q}(\alpha^{2m} \beta^{2n}) = \left(\sum_{|d| \leq R_{k,l}} [\alpha^{m+n+dw}] \right) (-1)^n \pi_{k,l} \bmod p. \quad \square$$

Corollary 15.1. *If -1 is a square in $\mathbb{F}_q$, then $\psi_{b,q}$ and $\psi_{r,q}$ are related in the following manner: for all $k, l \in \mathbb{N}$,*

$$\psi_{b,q}(\alpha^k \beta^l) = \begin{cases} \psi_{r,q}(\alpha^k \beta^l) & \text{if } l \not\equiv 2 \bmod 4 \\ -\psi_{r,q}(\alpha^k \beta^l) & \text{if } l \equiv 2 \bmod 4. \end{cases}$$

5.1 An explicit formula for $\psi_{b,q}$ when $\left(\frac{-1}{q}\right) = -1$ via complex summations

We conclude our analysis by giving an explicit formula for $\psi_{b,q}$ when $\left(\frac{-1}{q}\right) = -1$. We may think of $\mathbb{F}_q$ as being embedded in the quadratic extension $\mathbb{C}_{b,q}$, and so $\psi_{b,q}(\alpha^k \beta^l)$ might be identified as

$$\psi_{b,q}(\alpha^k \beta^l) 1_{\mathbb{D}} = \sum_{[x,y] \in S_{b,q}} x^k y^l 1_{\mathbb{D}}. \quad (18)$$

From identity (7), if $z = x1_{\mathbb{D}} + yi \in \mathbb{C}_{b,q}$ then

$$x1_{\mathbb{D}} = \frac{1}{2}(z + z^*) \quad \text{and} \quad y1_{\mathbb{D}} = -\frac{1}{2}(z - z^*)i.$$

Now any $z \in S_{b,q}$ satisfies $z^{-1} = z^*$, so for any such z we may write

$$x1_{\mathbb{D}} = \frac{1}{2}(z + z^{-1}) \quad \text{and} \quad y1_{\mathbb{D}} = -\frac{1}{2}(z - z^{-1})i.$$

This allows us to find a formula for $\psi_{b,q}(\alpha^k \beta^l) 1_{\mathbb{D}}$ in (18) as an algebraic function in z instead of a function in x and y given by

$$\begin{aligned} \psi_{b,q}(\alpha^k \beta^l) 1_{\mathbb{D}} &= \sum_{[x,y] \in S_{b,q}} x^k y^l 1_{\mathbb{D}} \\ &= \sum_{[x,y] \in S_{b,q}} (x1_{\mathbb{D}})^k (y1_{\mathbb{D}})^l \\ &= \sum_{z \in S_{b,q}} \left(\frac{1}{2}(z + z^{-1}) \right)^k \left(-\frac{1}{2}(z - z^{-1})i \right)^l. \end{aligned} \quad (19)$$

Therefore, a formula for $\psi_{b,q}(\alpha^k \beta^l)$ can be found in reference to summations of integer powers of complex numbers over $S_{b,q}$. Note that this is akin to an integration of the form $\int_{S^1} e^{in\theta} d\theta$ from classical Fourier analysis.

We are now ready to present a formula for $\psi_{b,q}(\alpha^k \beta^l)$ which is parallel to the other Fourier summation formulas that we have already obtained.

Theorem 16. *In the blue geometry over $\mathbb{F}_q$ when -1 is not a square, define $w \equiv \frac{1}{2}|S_{b,q}| = \frac{1}{2}(q+1)$ and for natural numbers k and l , let $R_{k,l} \equiv \left\lfloor \frac{k+l}{|S_{b,q}|} \right\rfloor$. We have that*

$$\psi_{b,q}(\alpha^k \beta^l) = \begin{cases} \left(\sum_{|d| \leq R_{k,l}} [\alpha^{m+n+dw}] \right) (-1)^n \pi_{k,l} \bmod p & \text{if } k = 2m, l = 2n \\ 0 & \text{otherwise.} \end{cases}$$

Proof. By the embedding $\psi_{b,q}(\alpha^k \beta^l) \mapsto \psi_{b,q}(\alpha^k \beta^l) 1_{\mathbb{D}}$, from (19) we have that

$$\begin{aligned} \psi_{b,q}(\alpha^k \beta^l) 1_{\mathbb{D}} &= \frac{(-1)^l}{2^{k+l}} i^l \sum_{z \in S_{b,q}} (z + z^{-1})^k (z - z^{-1})^l \\ &= \frac{(-1)^l}{2^{k+l}} i^l \sum_{z \in S_{b,q}} \left(\sum_{s=0}^k \binom{k}{s} z^{2s-k} \right) \left(\sum_{t=0}^l (-1)^{l-t} \binom{l}{t} z^{2t-l} \right) \\ &= \frac{1}{2^{k+l}} i^l \sum_{s=0}^k \sum_{t=0}^l (-1)^t \binom{k}{s} \binom{l}{t} \sum_{z \in S_{b,q}} z^{2(s+t)-(k+l)}. \end{aligned} \quad (20)$$

In the third equality above, we interchanged the summation order.

By Proposition 3, $S_{b,q}$ is a cyclic group of order $q+1$, and so by Lemma 4, it follows that

$$\sum_{z \in S_{b,q}} z^{2(s+t)-(k+l)} = \begin{cases} 1_{\mathbb{D}} & \text{if } q+1 \text{ divides } 2(s+t) - (k+l) \\ 0 & \text{otherwise} \end{cases}$$

and so the terms in (20) will vanish except for those s and t for which the equation

$$2(s+t) = k+l + d(q+1) \quad (21)$$

has solutions for d . Similar to the proof of Theorem 13, the case $k+l$ is odd gives $\psi_{b,q}(\alpha^k \beta^l) = 0$ and if $k+l = 2N$ for some integer N , then (21) has solutions for d precisely when $|d| \leq R_{k,l}$.

Each d where $|d| \leq R_{k,l}$ then contributes to the specific equation $s+t = N+dw$ where $w = \frac{1}{2} |S_{b,q}|$. We can simplify (20) to

$$\psi_{b,q}(\alpha^k \beta^l) 1_{\mathbb{D}} = \frac{1}{2^{2N}} i^l \sum_{|d| \leq R_{k,l}} \sum_{s=0}^k (-1)^{N+dw-s} \binom{k}{s} \binom{l}{N+dw-s} 1_{\mathbb{D}}. \quad (22)$$

The same analysis pertains as in the red geometry: we have that

$$\sum_{s=0}^k (-1)^{N+dw-s} \binom{k}{s} \binom{l}{N+dw-s} = [\alpha^{N+dw}] \left((1+\alpha)^k (1-\alpha)^l \right)$$

so if k and l are both odd, then the right-hand side of (22) vanishes and if $k = 2m$ and $l = 2n$, then we may rewrite (22) as

$$\begin{aligned} \psi_{b,q}(\alpha^k \beta^l) 1_{\mathbb{D}} &= \frac{1}{2^{2(m+n)}} i^{2n} \sum_{|d| \leq R_{k,l}} ([\alpha^{m+n+dw}] ((1+\alpha)^{2m} (1-\alpha)^{2n})) 1_{\mathbb{D}} \\ &= \left(\sum_{d \leq R_{k,l}} [\alpha^{m+n+dw}] (-1)^n \left(\left(\frac{1+\alpha}{2} \right)^{2m} \left(\frac{1-\alpha}{2} \right)^{2n} \right) \right) 1_{\mathbb{D}}. \end{aligned}$$

It then follows that

$$\begin{aligned} \psi_{b,q}(\alpha^k \beta^l) &= \sum_{d \leq R_{k,l}} [\alpha^{m+n+dw}] (-1)^n \left(\left(\frac{1+\alpha}{2} \right)^{2m} \left(\frac{1-\alpha}{2} \right)^{2n} \right) 1_q \\ &= \left(\sum_{|d| \leq R_{k,l}} [\alpha^{m+n+dw}] \right) (-1)^n \pi_{2m,2n} \bmod p. \end{aligned} \quad \square$$

5.2 A combined formula for $\psi_{b,q}$

Both formulas for $\psi_{b,q}$ are exactly the same except for the geometric constant w and consequently $R_{k,l}$. However, we can combine both formulas into one to make it similar to the formula for $\psi_{g,q}$ and $\psi_{r,q}$ established in Theorem 12 and 13.

Theorem 17 (A unified formula for $\psi_{b,q}$). *In the blue geometry over a general finite field $\mathbb{F}_q$ of size $q = p^r$ for some prime $p > 2$ and positive integer r , define a geometric constant*

$$w \equiv \frac{1}{2} |S_{b,q}| = \frac{1}{2} \left(q - \left(\frac{-1}{q} \right) \right).$$

For any natural numbers k and l , let $R_{k,l} \equiv \left\lfloor \frac{k+l}{|S_{b,q}|} \right\rfloor$. We have that

$$\psi_{b,q}(\alpha^k \beta^l) = \begin{cases} \left(\sum_{|d| \leq R_{k,l}} [\alpha^{m+n+dw}] \right) (-1)^n \pi_{k,l} \bmod p & \text{if } k = 2m, l = 2n \\ 0 & \text{otherwise.} \end{cases}$$

Note that this formula is strikingly similar to the formula of $\psi_{r,q}$, with the only difference being the presence of the factor $(-1)^n$. Moreover, we have the same analogue as in Theorem 14.

Theorem 18. *In the blue geometry over a field $\mathbb{F}_q$, for any $k, l \in \mathbb{N}$ such that $0 \leq k + l < q - 1$, the blue Fourier summation functional has the form*

$$\psi_{b,q}(\alpha^k \beta^l) = \begin{cases} \Omega(m, n) \bmod p & \text{if } k = 2m \text{ and } l = 2n \\ 0 & \text{otherwise.} \end{cases}$$

Proof. If either k or l is odd, then $\psi_{b,q}(\alpha^k \beta^l) = 0$ regardless of whether -1 is a square or not in $\mathbb{F}_q$. Moreover, if $0 \leq k + l < q - 1$ then by Theorem 17, $R_{k,l} = \left\lfloor \frac{k+l}{|S_{b,q}|} \right\rfloor = 0$ and thus if $k = 2m$ and $l = 2n$, by Theorem 7 we consequently have

$$\psi_{b,q}(\alpha^k \beta^l) = [\alpha^{m+n}] (-1)^n \pi_{2m,2n} = \Omega(m, n) \bmod p. \quad \square$$

We conclude our analysis of the derivation of the formula for the Fourier summation functional $\psi_{b,q}$ by highlighting the similarity of the formula for $\psi_{b,q}(\alpha^k \beta^l)$ when $0 \leq k + l < q - 1$ above with the characteristic 0 integration formula by Baker and Folland. We restate the result below for convenience, by renaming the polynumber to $\alpha^k \beta^l$ to match with our notation:

$$\int_{S_2} \alpha^k \beta^l d\mu = \begin{cases} \Omega(m, n) & \text{if } k = 2m \text{ and } l = 2n \\ 0 & \text{otherwise.} \end{cases}$$

A pleasant feature of the analysis that we have presented here lies in its flexibility to extend to general fields of characteristic zero. In a further paper, we will show that the traditional unit circle integration theory over the field of real numbers, that utilizes transcendental functions and infinite processes and limits, can be replaced with a purely algebraic approach in terms of a circular integral functional, and the pleasant connection to the super Catalan numbers suggests that there is something combinatorial about this theory of integration. With this algebraic approach, the connection between algebraic combinatorics and harmonic analysis can be better appreciated, and the characteristic zero and finite field situations can be viewed more uniformly.

6 Some examples of the computation of $\psi_{c,q}$

Here are some examples of computing $\psi_{c,q}$.

Example 5. Let $q = 17$. From Example 3, we know that $\psi_{r,17}(\alpha^6\beta^4) = 3$. Now, using the result in Theorem 13, we confirm this:

$$\psi_{r,17}(\alpha^6\beta^4) = (-1)^2 \Omega(3, 2) \bmod 17 = \frac{3}{256} \bmod 17 = 3.$$

Now if $k = 40$ and $l = 24$, then $R_{k,l} = \left\lfloor \frac{k+l}{q-1} \right\rfloor = 4$ and therefore

$$\begin{aligned} \psi_{r,17}(\alpha^{40}\beta^{24}) &= \sum_{|d| \leq 4} [\alpha^{32+8d}] \pi_{40,24} \bmod 17 \\ &= ([\alpha^0] + [\alpha^8] + [\alpha^{16}] + [\alpha^{24}] + [\alpha^{32}] + [\alpha^{40}] + [\alpha^{48}] + [\alpha^{56}] + [\alpha^{64}]) \pi_{40,24}. \end{aligned}$$

Through inspection,

$$\begin{aligned} [\alpha^0] \pi_{40,24} &= [\alpha^{64}] \pi_{40,24} = \frac{1}{2^{64}} \\ [\alpha^8] \pi_{40,24} &= [\alpha^{56}] \pi_{40,24} = \frac{90744}{2^{64}} \\ [\alpha^{16}] \pi_{40,24} &= [\alpha^{48}] \pi_{40,24} = -\frac{15\,156\,452}{2^{64}} \\ [\alpha^{24}] \pi_{40,24} &= [\alpha^{40}] \pi_{40,24} = -\frac{264\,053\,432}{2^{64}} \\ [\alpha^{32}] \pi_{40,24} &= \frac{1650\,887\,238}{2^{64}} \end{aligned}$$

and thus

$$\begin{aligned} \psi_{r,17}(\alpha^{40}\beta^{24}) &= \left(2 \left(\frac{1}{2^{64}} + \frac{90744}{2^{64}} - \frac{15\,156\,452}{2^{64}} - \frac{264\,053\,432}{2^{64}} \right) + \frac{1650\,887\,238}{2^{64}} \right) \bmod 17 \\ &= \frac{33345}{2^{49}} \bmod 17 = 4. \end{aligned}$$

Example 6. Let $q = 27 = 3^3$. From Example 4, $\left(\frac{-1}{3}\right) = -1$ and $\psi_{b,27}(\alpha^2) = 2$ which agrees with Theorem 16. Moreover, if $k = l = 26$, then $w = \frac{1}{2}(q+1) = 14$ and $R_{k,l} = \left\lfloor \frac{k+l}{q+1} \right\rfloor = 1$. Therefore

$$\begin{aligned} \psi_{b,27}(\alpha^{26}\beta^{26}) &= \sum_{|d| \leq 1} [\alpha^{26+14d}] (-1)^{13} \pi_{26,26} \bmod 3 \\ &= -([\alpha^{12}] + [\alpha^{26}] + [\alpha^{40}]) \pi_{26,26} \bmod 3 \\ &= -\left(\frac{230\,230}{2^{52}} - \frac{10\,400\,600}{2^{52}} + \frac{230\,230}{2^{52}} \right) \bmod 3 = 0. \end{aligned}$$

7 Fourier Summation Program via Periodicity

Since any polynumber $\pi \in \text{Pol}_2(\mathbb{F}_q)$ admits a unique decomposition $\pi_1 + \pi_2$ where $\pi_1 \in \mathcal{P}_q$, $\pi_2 \in \ker(\varepsilon)$ by Proposition 1 and $\psi_{c,q}(\pi) = \psi_{c,q}(\pi_1)$ from the proof of Theorem 11, there ought to be a simplification of the formulas for $\psi_{c,q}$ especially if the degree of the input polynumber π is large.

In this final section, we obtain a further simplification of the formulas for $\psi_{r,q}$ and $\psi_{b,q}$ by reducing the degree of the input polynumber π so that it falls in the principal subspace $\mathcal{P}_q$.

Theorem 19. *Let $\psi_{c,q}$ be the Fourier summation functional in any of the three geometries. For all natural numbers k, l, r , and s , we have that*

$$\psi_{c,q}(\alpha^{k+rq}\beta^{l+sq}) = \psi_{c,q}(\alpha^{k+r}\beta^{l+s}).$$

Proof. The polynumber $\pi = \alpha^{k+rq}\beta^{l+sq} - \alpha^{k+r}\beta^{l+s}$ evaluates to the zero function since

$$\varepsilon(\pi)(x, y) = x^{k+rq}y^{l+sq} - x^{k+r}y^{l+s} = x^k y^l ((x^r y^s)^q - x^r y^s) = 0.$$

The conclusion follows by the Locality property. □

To illustrate the Periodicity property, we consider again the case $q = 13$ in the blue geometry and gather a table of values of $\psi_{b,13}(\alpha^{2m}\beta^{2n})$ for various m and n , since $\psi_{b,13}(\alpha^k\beta^l) = 0$ if either k or l is odd. By Theorem 19, all the information required to determine $\psi_{b,13}(\alpha^{2m}\beta^{2n})$ is contained in the following finite list of values of $\psi_{b,13}(\alpha^{2m}\beta^{2n})$ for $0 \leq m, n \leq 6$.

$\psi_{b,13}(\alpha^{2m}\beta^{2n}) : m \setminus n$	0	1	2	3	4	5	6
0	1	7	2	6	2	7	3
1	7	5	9	4	8	4	9
2	2	9	5	9	4	8	4
3	6	4	9	5	9	4	8
4	2	8	4	9	5	9	4
5	7	4	8	4	9	5	9
6	3	9	4	8	4	9	5

Table 6: All the information to determine $\psi_{b,13}(\alpha^{2m}\beta^{2n})$.

For example, $\psi_{b,13}(\alpha^{100}\beta^{200}) = \psi_{b,13}(\alpha^{16}\beta^{20}) = \psi_{b,13}(\alpha^4\beta^8) = 4$, where we apply Theorem 19 above twice: the first one with $(k, l, r, s) = (9, 5, 7, 15)$ and the second one with $(k, l, r, s) = (3, 7, 1, 1)$.

In general, in the red and blue geometries, it is sufficient to find all the values of $\psi_{c,q}(\alpha^{2m}\beta^{2n})$ where $0 \leq m, n \leq \frac{1}{2}(q-1)$ since $\psi_{c,q}(\alpha^k\beta^l) = 0$ if either k or l is odd by Theorem 13 and 17. We first consider an explicit formula for $\psi_{r,q}$ and $\psi_{b,q}$ evaluated at $\alpha^{2m}\beta^{2n} \in \mathcal{P}_q$ in the following proposition.

Proposition 5. *Suppose $\mathbb{F}_q$ is a finite field with $q = p^r$ elements for some prime $p > 2$ and positive integer r .*

1. In the red geometry, let $w = \frac{1}{2}(q-1)$. If $0 \leq m, n \leq w$, then

$$\psi_{r,q}(\alpha^{2m}\beta^{2n}) = \left((-1)^n \Omega(m, n) + 2[\alpha^{m+n+w}] \pi_{2m,2n} + \frac{2}{4^{m+n}} \delta_{m+n,2w} \right) \bmod p.$$

2. In the blue geometry where $\left(\frac{-1}{q}\right) = 1$, let $w = \frac{1}{2}(q-1)$. If $0 \leq m, n \leq w$, then

$$\psi_{b,q}(\alpha^{2m}\beta^{2n}) = \left(\Omega(m, n) + 2[\alpha^{m+n+w}] (-1)^n \pi_{2m,2n} + \frac{2}{4^{m+n}} (-1)^n \delta_{m+n,2w} \right) \bmod p.$$

3. In the blue geometry where $\left(\frac{-1}{q}\right) = -1$, let $w = \frac{1}{2}(q+1)$. If $0 \leq m, n \leq w-1$, then

$$\psi_{b,q}(\alpha^{2m}\beta^{2n}) = (\Omega(m, n) + 2[\alpha^{m+n+w}] (-1)^n \pi_{2m,2n}) \bmod p.$$

Proof. In the red geometry, if $0 \leq m+n < w$, it follows from Theorem 13 that $R_{k,l} = \left\lfloor \frac{k+l}{q-1} \right\rfloor = 0$, so only the central term $(-1)^n \Omega(m, n) \bmod p$ will appear in the formula. If $w \leq m+n < 2w$, then $R_{k,l} = 1$ which gives us two additional terms $[\alpha^{m+n+w}] \pi_{2m,2n} \bmod p$ and $[\alpha^{m+n-w}] \pi_{2m,2n} \bmod p$ in the formula of $\psi_{r,q}$ and they are equal by Lemma 2. Finally, if $m+n = 2w$, we have two further additional terms which are equal, namely $[\alpha^{m+n+2w}] \pi_{2m,2n} \bmod p$ and $[\alpha^{m+n-2w}] \pi_{2m,2n} \bmod p$. Since

$$[\alpha^{m+n+2w}] \pi_{2m,2n} \bmod p = [\alpha^{2m+2n}] \left(\frac{1+\alpha}{2} \right)^{2m} \left(\frac{1-\alpha}{2} \right)^{2n} \bmod p = \frac{1}{4^{m+n}} \delta_{m+n,2w} \bmod p$$

the conclusion follows.

The second identity follows by the result of Theorem 15 that

$$\psi_{b,q}(\alpha^{2m}\beta^{2n}) = (-1)^n \psi_{r,q}(\alpha^{2m}\beta^{2n})$$

whenever -1 is a square in $\mathbb{F}_q$. The third identity is done similarly to the proof of the red geometry case. $\square$

The following result is a general program for calculating $\psi_{r,q}(\alpha^{2m}\beta^{2n})$ and $\psi_{b,q}(\alpha^{2m}\beta^{2n})$ that reduces the computational effort.

Theorem 20 (Fourier Summation Program). *Suppose $\mathbb{F}_q$ is a finite field with $q = p^r$ elements for some prime $p > 2$ and positive integer r . Given any natural numbers m and n , perform the following procedure:*

1. Set $m_0 = 2m$ and $n_0 = 2n$.

2. For each $k \in \mathbb{N}$, identify four integers Q_k, R_k, S_k , and T_k such that Q_k and S_k are the largest integers satisfying

$$m_k = Q_k q + R_k \text{ and } n_k = S_k q + T_k$$

for some natural numbers R_k and T_k .

3. Set $m_{k+1} = Q_k + R_k$ and $n_{k+1} = S_k + T_k$. If $m_{k+1} \geq q$ or $n_{k+1} \geq q$, then repeat step 2. If not, then set $M = m_{k+1}$ and $N = n_{k+1}$ and stop the procedure.

We have that

1. The procedure will halt;
2. $M = 2m^*$ and $N = 2n^*$ for some $m^*, n^* \in \mathbb{N}$; and
3. In the red geometry,

$$\psi_{r,q}(\alpha^{2m}\beta^{2n}) = \left((-1)^{n^*} \Omega(m^*, n^*) + 2[\alpha^{m^*+n^*+\frac{q-1}{2}}] \pi_{2m^*, 2n^*} + \frac{2}{4^{m^*+n^*}} \delta_{m^*+n^*, q-1} \right) \bmod p.$$

In the blue geometry, if -1 is a square in $\mathbb{F}_q$ then

$$\psi_{b,q}(\alpha^{2m}\beta^{2n}) = \left(\Omega(m^*, n^*) + 2(-1)^{n^*} [\alpha^{m^*+n^*+\frac{q-1}{2}}] \pi_{2m^*, 2n^*} + \frac{2(-1)^{n^*}}{4^{m^*+n^*}} \delta_{m^*+n^*, q-1} \right) \bmod p$$

and if -1 is not a square in $\mathbb{F}_q$ then

$$\psi_{b,q}(\alpha^{2m}\beta^{2n}) = \left(\Omega(m^*, n^*) + 2[\alpha^{m^*+n^*+\frac{q+1}{2}}] (-1)^{n^*} \pi_{2m^*, 2n^*} \right) \bmod p.$$

Proof. If the procedure does not halt, then the sequence (m_k) is a strictly decreasing sequence of natural numbers since

$$m_k = Q_k q + R_k > Q_k + R_k = m_{k+1}$$

by construction, which is impossible.

Now, since $m_0 = Q_0 q + R_0$ is even and q is odd, it follows that Q_0 and R_0 must both be odd or even, so $m_1 = Q_0 + R_0$ is even. By a similar argument, $n_1 = S_0 + T_0$ is even. We can keep on going to show that m_k and n_k are even. Since the procedure halts, M and N are both even.

Finally, by construction and Theorem 19,

$$\psi_{c,q}(\alpha^{m_k}\beta^{n_k}) = \psi_{c,q}(\alpha^{Q_k q + R_k} \beta^{S_k q + T_k}) = \psi_{c,q}(\alpha^{Q_k + R_k} \beta^{S_k + T_k}) = \psi_{c,q}(\alpha^{m_{k+1}} \beta^{n_{k+1}})$$

which shows that

$$\psi_{c,q}(\alpha^{2m}\beta^{2n}) = \psi_{c,q}(\alpha^{m_0}\beta^{n_0}) = \psi_{c,q}(\alpha^M \beta^N) = \psi_{c,q}(\alpha^{2m^*} \beta^{2n^*}).$$

Since $0 \leq m^*, n^* \leq \frac{1}{2}(q-1)$, we can apply Proposition 5 to prove the last assertion. $\square$

We close this section with an example.

Example 7. Suppose $q = 13$ and we wish to compute $\psi_{b,13}(\alpha^{1000}\beta^{600})$. If we were to use the formula for $\psi_{b,q}$ in Theorem 17, we need to find particular coefficients of the circular polynumber $\pi_{1000,600}$ of degree 1600. This is of course computationally expensive. However, with the aid of Theorem 20, we can perform the Fourier Summation Program applied to $m_0 = 1000$, $n_0 = 600$, and $q = 13$ to get

$$(m_1, n_1) = (88, 48) \quad (m_2, n_2) = (16, 12) \quad (m_3, n_3) = (4, 12).$$

Since now $m_3, n_3 < q$, the procedure halts. Therefore

$$\begin{aligned} \psi_{b,13}(\alpha^{1000}\beta^{600}) &= \psi_{b,13}(\alpha^4\beta^{12}) \\ &= \left(\Omega(2, 6) + 2(-1)^6 [\alpha^{14}] \left(\frac{1+\alpha}{2} \right)^4 \left(\frac{1-\alpha}{2} \right)^{12} \right) \bmod 13 \\ &= 9. \end{aligned}$$

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